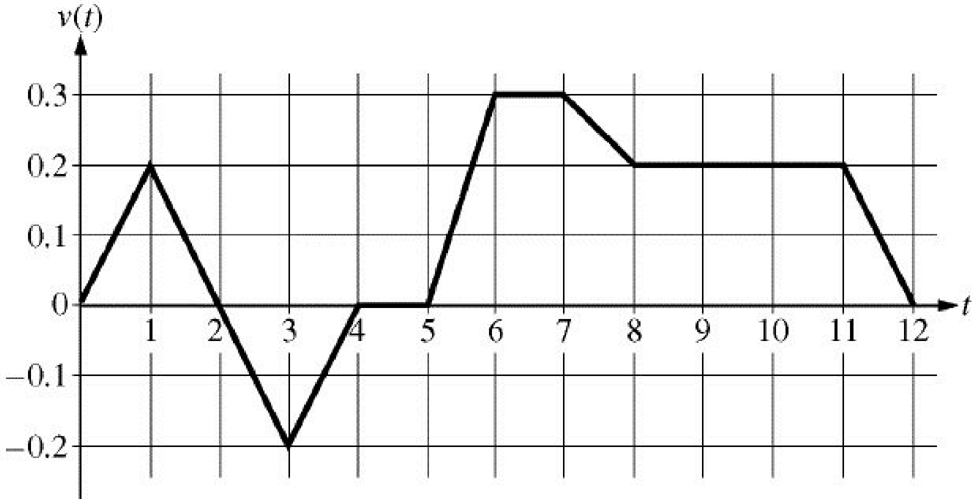
8-2b



Caren rides her bicycle along a straight road from home to school, starting at home at time *t*=0 minutes and arriving at school at time *t*=12 minutes. During the time interval $0 \le t \le 12$ minutes, her velocity *v(t)* , in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

1. Larry also rides his bicycle along a straight road from home to school in 12 minutes. His velocity is modeled by the function w given by w(t)=π/15sin(π/12t) , where w(t) is in miles per minute for $0 \le t \le 12$ minutes. Who lives closer to school: Caren or Larry? Show the work that leads to your answer.

Part D

1 point is earned for integral

1 point is earned for value of larry's distance

1 point is earned for value of caren's distane and conclusion

$\int_0^{12} w(t)dt = 1.6$; Larry lives 1.6 miles fr om school.

$\int_0^{12} v(t)dt = 1.4$; Caren lives 1.4 miles fr om school.

Therefore, Caren lives closer to school.

0	1	2	3

The students response earns three of the following points

1 point is earned for integral

8-2b

1 point is earned for value of larry's distance

1 point is earned for value of caren's distane and conclusion

$\int_0^{12} w(t)dt = 1.6$; Larry lives 1.6 miles fr om school.

$\int_0^{12} v(t)dt = 1.4$; Caren lives 1.4 miles fr om school.

Therefore, Caren lives closer to school.

8-2b

2.

Using correct units, explain the meaning of $\int_0^{12} |v(t)|dt$ in terms of Caren's trip. Find the value of $\int_0^{12} |v(t)|dt$.

Part B

1 point is earned for meaning of integral

1 point is earned for value of integral

$\int_0^{12} \left|v(t)\right| dt$ is the total distance, in miles, that Caren rode during the 12 minutes fr om t=0 to t=12.

$$\int_0^{12} \left|v(t)\right| dt = \int_0^2 v(t)dt - \int_2^4 v(t)dt + \int_4^{12} v(t)dt$$

=0.2+0.2+1.4=1.8 miles

0	1	2

The student response earns two of the following points:

1 point is earned for meaning of integral

1 point is earned for value of integral

$\int_0^{12} \left|v(t)\right| dt$ is the total distance, in miles, that Caren rode during the 12 minutes fr om t=0 to t=12.

$$\int_0^{12} \left|v(t)\right| dt = \int_0^2 v(t)dt - \int_2^4 v(t)dt + \int_4^{12} v(t)dt$$

=0.2+0.2+1.4=1.8 miles

8-2b

3. A particle moves along the x-axis so that its velocity at time t, $0 \le t \le 5$, is given by $v(t) = 3(t-1)(t-3)$. At time $t = 2$, the position of the particle is $x(2) = 0$.

(a) Find the minimum acceleration of the particle.

(b) Find the total distance traveled by the particle.

(c) Find the average velocity of the particle over the interval $0 \le t \le 5$.

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$v(t) = 3t^2 - 12t + 9$

$a(t) = 6t - 12$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a.

0	1	2	3

The student response earns all of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$v(t) = 3t^2 - 12t + 9$

$a(t) = 6t - 12$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a.

Part B

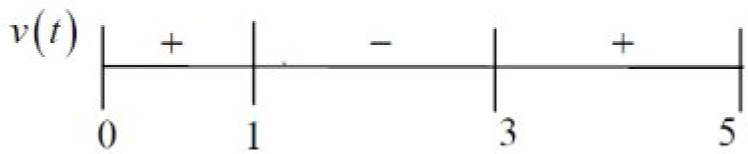
2 points are earned for integral expression

8-2b

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$d=\int_0^1(3t^2-12t+9)dt-\int_1^3(3t^2-12t+9)dt+\int_3^5(3t^2-12t+9)dt$$
$$=\left[t^3-6t^2+9t\right]_0^1-\left[t^3-6t^2+9t\right]_1^3+\left[t^3-6t^2+9t\right]_3^5$$
$$=4-(-4)+20=28$$

or

Method 2

$$x(t)=t^3-6t^2+9t-2$$
$$\left[(\text{or})\ x(t)=t^3-6t^2+9t+C\right]$$

$x(0)$ = -2

$x(1)$ = 2

$x(3)$ = -2

$x(5)$ = 18

Total distance = 4 + 4 + 20 = 28

0	1	2	3	4 ✓

The student response earns all of the following points:

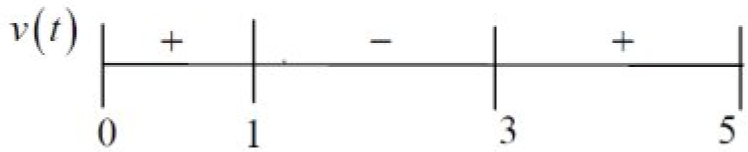
2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

8-2b

Method 1



$$d=\int_0^1(3t^2-12t+9)dt-\int_1^3(3t^2-12t+9)dt+\int_3^5(3t^2-12t+9)dt$$
$$=\left[t^3-6t^2+9t\right]_0^1-\left[t^3-6t^2+9t\right]_1^3+\left[t^3-6t^2+9t\right]_3^5$$
$$=4-(-4)+20=28$$

or

Method 2

$x(t)=t^3-6t^2+9t-2$

$\left[(\text{or})\ x(t)=t^3-6t^2+9t+C\right]$

$x(0) = -2$

$x(1) = 2$

$x(3) = -2$

$x(5) = 18$

Total distance = 4 + 4 + 20 = 28

Part C

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\frac{\int_0^5\left(3t^2-12t+9\right)dt}{5-0}$$
$$=\frac{1}{5}\left[t^3-6t^2+9t\right]_0^5=\frac{1}{5}(20)=4$$

or

8-2b

$\boxed{\text{Method 2}}$

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

$\boxed{\text{Method 1}}$

$$\frac{\int_0^5 \left(3t^2-12t+9\right) dt}{5-0}$$

$$= \frac{1}{5}\left[t^3 - 6t^2 + 9t\right]_0^5 = \frac{1}{5}(20) = 4$$

or

$\boxed{\text{Method 2}}$

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$

8-2b

4. A particle moves along the x-axis so that its velocity at time t , $0 \le t \le 5$, is given by $v(t) = 3(t-1)(t-3)$. At time $t = 2$, the position of the particle is $x(2) = 0$.

Find the total distance traveled by the particle.

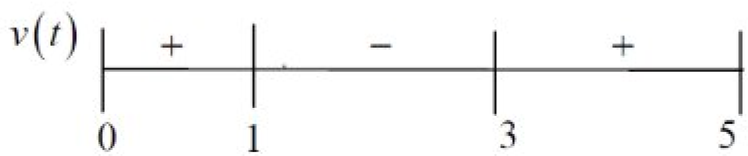
Part B

2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$d = \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt$$

$$= \left[t^3 - 6t^2 + 9t\right]_0^1 - \left[t^3 - 6t^2 + 9t\right]_1^3 + \left[t^3 - 6t^2 + 9t\right]_3^5$$

$$= 4 - (-4) + 20 = 28$$

Method 2

$x(t) = t^3 - 6t^2 + 9t - 2$

$\left[(or)x(t) = t^3 - 6t^2 + 9t + C\right]$

x(0) = -2

x(1) = 2

x(3) = -2

x(5) = 18

Total distance = 4 + 4 + 20 = 28

0	1	2	3	4

The student response earns four of the following points:

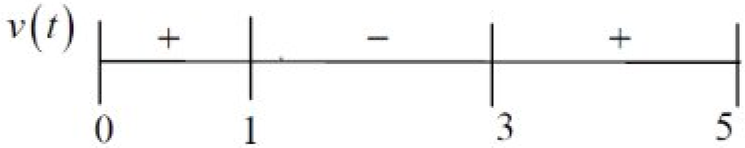
8-2b

2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$d=\int_0^1 (3t^2-12t+9)dt-\int_1^3 (3t^2-12t+9)dt+\int_3^5 (3t^2-12t+9)dt$$

$$=\left[t^3-6t^2+9t\right]_0^1-\left[t^3-6t^2+9t\right]_1^3+\left[t^3-6t^2+9t\right]_3^5$$

$$=4-(-4)+20=28$$

Method 2

$x(t)=t^3-6t^2+9t-2$

$\left[(or)x(t)=t^3-6t^2+9t+C\right]$

x(0) = -2

x(1) = 2

x(3) = -2

x(5) = 18

Total distance = 4 + 4 + 20 = 28

8-2b

5. Two particles move along the x-axis. For $0 \le t \le 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.

Write, but do not evaluate, an expression for the average distance between the two particles on the interval $1 \le t \le 3$.

Part D

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2}\int_1^3 |p(t) - r(t)|dt$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2}\int_1^3 |p(t) - r(t)|dt$$

8-2b

6.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t, measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t.

Using correct units, interpret the meaning of ∫060 $|v(t)|\ dt$ in the context of this problem. Approximate ∫060 $|v(t)|\ dt$ using a left Riemann sum with the sub intervals indicated by the data in the table.

Part B

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)|$ dt is the total distance, in meters, that Ben rides over the 60-second interval t=0 to t=60.

$\int_0^{60} |v(t)|$ dt ≈ 2.0 * 10+2.3(40-10)+2.5(60-40)=139 meters

0	1	2

The student response earns all of the following points:

1 point is earned for meaning of the integral

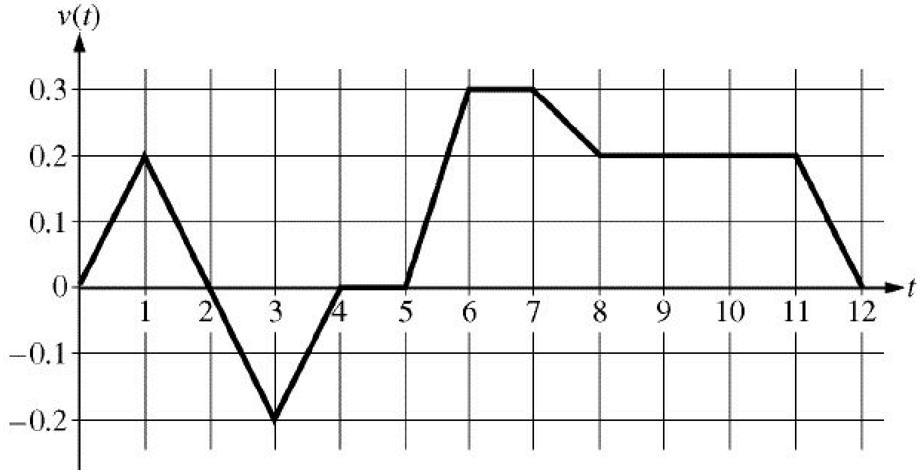
1 point is earned for the approximation

$\int_0^{60} |v(t)|$ dt is the total distance, in meters, that Ben rides over the 60-second interval t=0 to t=60.

$\int_0^{60} |v(t)|$ dt ≈ 2.0 * 10+2.3(40-10)+2.5(60-40)=139 meters

8-2b

7.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t = 0$ minutes and arriving at school at time $t = 12$ minutes. During the time interval $0 \le t \le 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

(a) Find the acceleration of Caren's bicycle at time $t = 7.5$ minutes. Indicate units of measure.

(b) Using correct units, explain the meaning of munderover $\int_0^{12} |v(t)|dt$ in terms of Caren's trip. Find the value of munderover $\int_0^{12} |v(t)|dt$.

(c) Shortly after leaving home, Caren realizes she left her calculus homework at home, and she returns to get it. At what time does she turn around to go back home? Give a reason for your answer.

(d) Larry also rides his bicycle along a straight road from home to school in 12 minutes. His velocity is modeled by the function w given by $w(t) = \frac{\pi}{15}\sin\left(\frac{\pi}{12}t\right)$, where $w(t)$ is in miles per minute for $0 \le t \le 12$ minutes. Who lives closer to school: Caren or Larry? Show the work that leads to your answer.

Part A

1 point is earned for the answer

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

8-2b

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

Part B

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)|dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)|dt = \int_0^2 v(t)dt - \int_2^4 v(t)dt + \int_4^{12} v(t)dt$$

$= 0.2 + 0.2 + 1.4 = 1.8$ miles

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)|dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)|dt = \int_0^2 v(t)dt - \int_2^4 v(t)dt + \int_4^{12} v(t)dt$$

$= 0.2 + 0.2 + 1.4 = 1.8$ miles

Part C

1 point is earned for the answer

8-2b

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative..

Part D

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$\int_0^{12} w(t)dt = 1.6$; Larry lives 1.6 miles from school.

$\int_0^{12} v(t)dt = 1.4$; Caren lives 1.4 miles from school.

Therefore, Caren lives closer to school.

0	1	2	3

The students response earns all of the following points:

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$\int_0^{12} w(t)dt = 1.6$; Larry lives 1.6 miles from school.

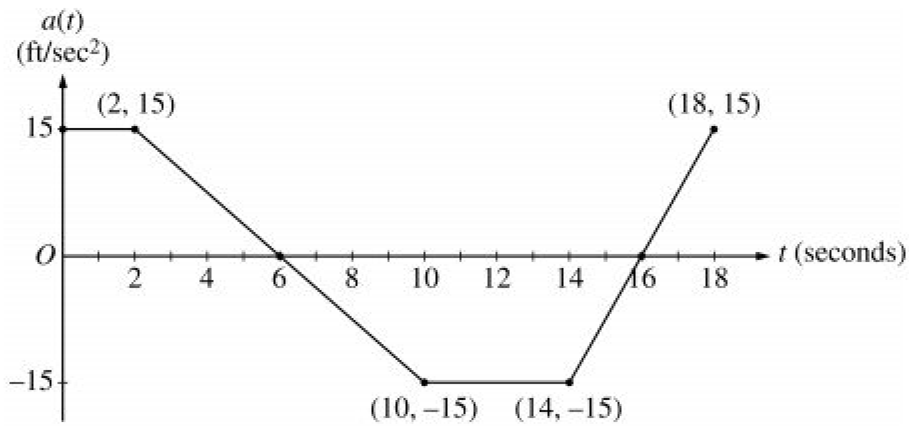
8-2b

$\int_0^{12} v(t)dt = 1.4$; Caren lives 1.4 miles from school.

Therefore, Caren lives closer to school.

8-2b

8.



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \le t \le 18$ seconds, the car's acceleration $a(t)$, in ft/sec^2, is the piecewise linear function defined by the graph above.

(a) Is the velocity of the car increasing at $t = 2$ seconds? Why or why not?

(b) At what time in the interval $0 \le t \le 18$, other than $t = 0$, is the velocity of the car 55 ft/sec? Why?

(c) On the time interval $0 \le t \le 18$, what is the car's absolute maximum velocity, in ft/sec, and at what time does it occur? Justify your answer.

(d) At what times in the interval $0 \le t \le 18$, if any, is the car's velocity equal to zero? Justify your answer.

Part A

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.

0	1

The student response earns all of the following points:

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.

Part B

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t)dt = 0$.

8-2b

0	1	2

The student response earns all of the following points:

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t)dt = 0.$

Part C

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$v(6) = 55 + \int_0^6 a(t)dt$$
$$= 55 + 2(15) + \tfrac{1}{2}(4)(15) = 115 > v(0)$$
$$\int_6^{18} a(t)dt < 0 \text{ so } v(18) < v(6)$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

8-2b

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$v(6) = 55 + \int_0^6 a(t)dt$$
$$= 55 + 2(15) + \tfrac{1}{2}(4)(15) = 115 > v(0)$$
$$\int_6^{18} a(t)dt < 0 \text{ so } v(18) < v(6)$$

Part D

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t)dt = 115 - 105 = 10 > 0.$$

0	1	2 ✓

The student response earns all of the following points:

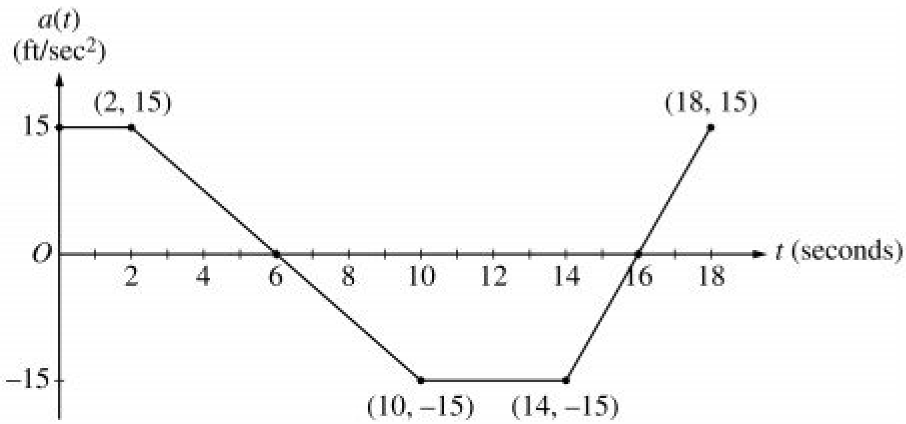
1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t)dt = 115 - 105 = 10 > 0.$$

8-2b



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \leq t \leq 18$ seconds, the car's acceleration $a(t)$, in ft/sec^2, is the piecewise linear function defined by the graph above

9. On the time interval $0 \leq t \leq 18$, what is the car's absolute maximum velocity, in ft/sec, and at what time does it occur? Justify your answer.

Part C

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifies $t = 6$ and $t = 18$ as candidates or indicates that v increases, decreases, then increases.

1 point is earned for eliminates $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$v(6) = 55 + \int_0^6 a(t)dt$$

$$= 55 + 2(15) + \tfrac{1}{2}(4)(15) = 115 > v(0)$$

$$\int_6^{18} a(t)dt < 0, so, v(18) < v(16)$$

0	1	2	3	4

The student response earns all of the following points:

8-2b

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifies $t = 6$ and $t = 18$ as candidates or indicates that v increases, decreases, then increases.

1 point is earned for eliminates $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$v(6) = 55 + \int_0^6 a(t)dt$$

$$= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0)$$

$$\int_6^{18} a(t)dt < 0, so, v(18) < v(16)$$

8-2b

10. At what time in the interval $0 \leq t \leq 18$, if any, is the car's velocity equal to zero? Justify your answer.

Part D

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v\,(16) = 115 + \int_{6}^{16} a\,(t)dt = 115 - 105 = 10 > 0.$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v\,(16) = 115 + \int_{6}^{16} a\,(t)dt = 115 - 105 = 10 > 0.$$

8-2b

11. At what time in the interval $0 \leq t \leq 18$, other than $t = 0$, is the velocity of the car 55 ft/sec? Why?

Part B

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because

$$v(12) - v(0) = \int_0^{12} a(t)dt = 0$$

0	1	2

The student response earns all of the following points:

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because

$$v(12) - v(0) = \int_0^{12} a(t)dt = 0$$

8-2b

12.

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	–20	–30	–20	–14	–10	0	10
$a(t)$ $\left(\text{ft}/\text{sec}^2\right)$	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \le t \le 60$ seconds, the car's velocity v, measured in feet per second, and acceleration a, measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

(a) Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)|dt$ in terms of the car's motion. Approximate $\int_{30}^{60} |v(t)|dt$ using a trapezoidal approximation with the three subintervals determined by the table.

(b) Using appropriate units, explain the meaning of $\int_{0}^{30} a(t)dt$ in terms of the car's motion. Find the exact value of $\int_{0}^{30} a(t)dt$.

(c) For $0 < t < 60$, must there be a time t when $v(t) = -5$? Justify your answer.

(d) For $0 < t < 60$, must there be a time t when $a(t) = 0$? Justify your answer.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)|dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)|dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

8-2b

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} |v(t)|dt$ is the distance in feet that the car travels from $t = 30$ sec to $t = 60$ sec.

Trapezoidal approximation for $\int_{30}^{60} |v(t)|dt$:

$$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185 \text{ ft}$$

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_{0}^{30} a(t)\, dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_{0}^{30} \mathrm{a}(t)\, dt = \int_{0}^{30} v'(t)\, dt = v(30) - v(0)$$

= -14 - (-20) = 6 ft/sec

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

8-2b

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t)\, dt$ is the car's change in velocity in ft/sec from $t = 0$ sec to $t = 30$ sec.

$$\int_0^{30} \mathrm{a}(t)\, dt = \int_0^{30} v'(t)\, dt = v(30) - v(0)$$

$= -14 - (-20) = 6$ ft/sec

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

Part C

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for $v(35) < -5 < v(50)$

1 point is earned for Yes; refers to IVT or hypotheses

Yes. Since $v(35) = -10 < -5 < 0 = v(50)$, the IVT guarantees a t in $(35, 50)$ so that $v(t) = -5$.

Part D

8-2b

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for $v(0) = v(25)$

1 point is earned for Yes; refers to MVT or hypotheses

Yes. Since $v(0) = v(25)$, the MVT guarantees a t in $(0,25)$ so that $a(t) = v'(t) = 0$.

8-2b

t (sec)	0	15	25	30	35	50	60
$v(t)$ (ft/sec)	−20	−30	−20	−14	−10	0	10
$a(t)$ (ft/sec^2)	1	5	2	1	2	4	2

A car travels on a straight track. During the time interval $0 \le t \le 60$ seconds, the car's velocity v, measured in feet per second, and acceleration a, measured in feet per second per second, are continuous functions. The table above shows selected values of these functions.

13. Using appropriate units, explain the meaning of $\int_0^{30} a(t)dt$ in terms of the car's motion. Find the exact value of $\int_0^{30} a(t)dt$.

Part B

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t)dt$ is the car's change in velocity in ft/sec from t=0 sec to t=30 sec.

$$\int_0^{30} a(t)dt = \int_0^{30} v\prime(t)dt = v(30) - v(0)$$
$$= -14 - (-20) = 6ft/\text{sec}$$

0	1	2

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_0^{30} a(t)dt$ is the car's change in velocity in ft/sec from t=0 sec to t=30 sec.

8-2b

$$\int_0^{30} a(t)dt = \int_0^{30} v\prime(t)dt = v(30) - v(0)$$
$$= -14 - (-20) = 6ft/\,\mathrm{sec}$$

Extra Point

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

0	1

The student response earns one of the following points:

1 point is earned for units in both (a) and (b)

Units of ft in (a) and ft/sec in (b)

8-2b

14. Using appropriate units, explain the meaning of $\int_{30}^{60} |v(t)| dt$ in terms of the car's motion. Approximate $\int_{30}^{60} |v(t)| dt$ using a trapezoidal approximation with the three sub intervals determined by the table.

Part A

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} \left| v(t) \right| dt$ is the distance in feet that the car travels from t=30 sec to t=60 sec.

Trapezoidal approximation for $\int_{30}^{60} \left| v(t) \right| dt$

$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185ft$

0	1	2

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the value

$\int_{30}^{60} \left| v(t) \right| dt$ is the distance in feet that the car travels from t=30 sec to t=60 sec.

Trapezoidal approximation for $\int_{30}^{60} \left| v(t) \right| dt$:

$A = \frac{1}{2}(14 + 10)5 + \frac{1}{2}(10)(15) + \frac{1}{2}(10)(10) = 185ft$

8-2b

For $0 \leq t \leq 6$, a particle is moving along the x-axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t)= 2\sin(e^{t/4})+1$. The acceleration of the particle is given by $a(t)= 1/2\, e^{t/4}\cos(e^{t/4})$ and $x(0)=2$.

15. For $0 \leq t \leq 6$ the particle changes direction exactly once. Find the position of the particle at that time.

Part D

1 point is earned for considering $v(t)=0$

1 point is earned for integral

1 point is earned for the answer

$v(t)=0$ when $t = 5.19552$. Let $b = 5.19552$

$v(t)$ changes sign from positive to negative at time $t=b$.

$$x(b) = 2 + \int_0^b v(t)dt = 14.134 \text{ or } 14.135$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $v(t)=0$

1 point is earned for integral

1 point is earned for the answer

$v(t)=0$ when $t = 5.19552$. Let $b = 5.19552$

$v(t)$ changes sign from positive to negative at time $t=b$.

$$x(b) = 2 + \int_0^b v(t)dt = 14.134 \text{ or } 14.135$$

8-2b

16. Find the average velocity of the particle for the time period $0 \leq t \leq 6$.

Part B

1 point is earned for integral

1 point is earned for the answer

Average velocity $= \frac{1}{6}\int_0^6 v(t)dt = 1.949$

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

Average velocity $= \frac{1}{6}\int_0^6 v(t)dt = 1.949$

8-2b

17. Find the total distance traveled by the particle from time $t = 0$ to $t = 6$.

Part C

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)|dt = 12.573$$

0	1	2 ✓

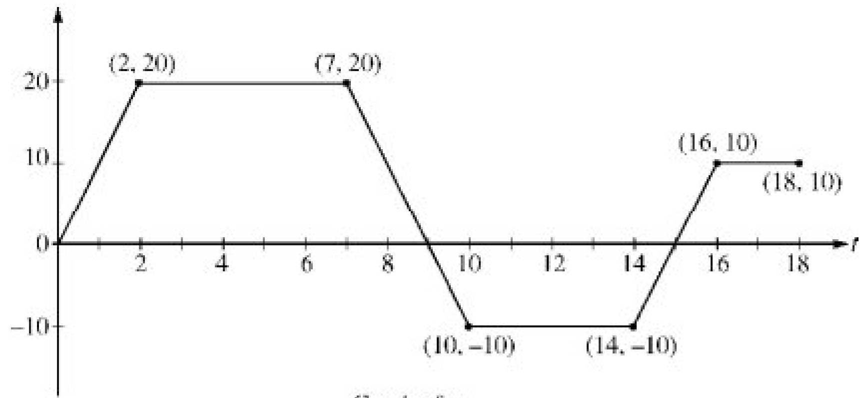
The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)|dt = 12.573$$

8-2b



Graph of v

A squirrel starts at building *A* at time t=0 and travels along a straight,horizontal wire connected to building *B*. For $0 \le t \le 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

18. Write expressions for the squirrel's acceleration $a(t)$, velocity $v(t)$, and distance $x(t)$ from building *A* that are valid for the time interval $7<t<10$.

Part D

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

$$For7 < t < 10, a(t) = \frac{20-(-10)}{7-10} = -10$$
$$v(t) = 20 - 10(t-7) = -10t + 90$$
$$x(7) = \frac{7+5}{2}.20 = 120$$
$$x(t) = x(7) + \int_7^t (-10u + 90)du$$
$$= 120 + (-5u^2 + 90u)\Big|_{u=7}^{u=t}$$
$$= -5t^2 + 90t - 265$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $a(t)$

1 point is earned for $v(t)$

8-2b

2 points are earned for $x(t)$

$$For 7 < t < 10, a(t) = \frac{20-(-10)}{7-10} = -10$$
$$v(t) = 20 - 10(t-7) = -10t + 90$$
$$x(7) = \frac{7+5}{2}.20 = 120$$
$$x(t) = x(7) + \int_7^t (-10u + 90)du$$
$$= 120 + (-5u^2 + 90u)\Big|_{u=7}^{u=t}$$
$$= -5t^2 + 90t - 265$$

19. Find the total distance the squirrel travels during the time interval $0 \le t \le 18$.

Part C

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)|dt = 140 + 50 + 25 = 215$

0	1

The student response earns one of the following points:

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)|dt = 140 + 50 + 25 = 215$

8-2b

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	−10	−8	−4	7

The velocity of a particle moving along the x-axis is modeled by a differentiable function v, where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position x=7 meters when t=0 seconds.

20. Suppose that the acceleration of the particle is positive for $0<t<8$ seconds. Explain why the position of the particle at t=8 seconds must be greater than x=30 meters.

Part D

1 point is earned for $v'(t)=a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t)=a(t) > 0$ for $0<t<8$, $v(t) \geq 3$ on this interval.

Therefore, $x(8)=x(0)+\int_0^8 v(t)dt \geq 7+8.3 > 30.$

0	1	2

The student response earns all of the following points:

1 point is earned for $v'(t)=a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t)=a(t) > 0$ for $0<t<8$, $v(t)$3 on this interval.

Therefore, $x(8)=x(0)+\int_0^8 v(t)dt \geq 7+8.3 > 30.$

8-2b

21. Using correct units, explain the meaning of $\int_{20}^{40} v(t)\,dt$ in the context of this problem. Use a trapezoidal sum with the three subintervals indicated by the data in the table to approximate $\int_{20}^{40} v(t)\,dt$.

Part B

1 point is earned units in (a) and (b)

1 point is earned for meaning of $\int_{20}^{40} v(t)dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t)dt$ is the particle's change in position in meters from time t=20 seconds to time t=40 seconds.

$$\int_{20}^{40} v(t)dt = \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

= -75 meters

0	1	2	3

The student response earns three of the following points:

1 point is earned units in (a) and (b)

1 point is earned for meaning of $\int_{20}^{40} v(t)dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t)dt$ is the particle's change in position in meters from time t=20 seconds to time t=40 seconds.

$$\int_{20}^{40} v(t)dt = \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

= -75 meters

8-2b

A particle moves along the x-axis so that its velocity v at time t, for $0 \leq t \leq 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

22. Find the average speed of the particle over the interval $0 \leq t \leq 2$.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2}\int_0^2 |v(t)|dt = 0.370 \text{ or } 0.371$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2}\int_0^2 |v(t)|dt = 0.370 \text{ or } 0.371$$

8-2b

23. Find the position of the particle at time t=2.

Part C

1 point is earned for $\int_0^2 \ln\left(u^2-3u+3\right)du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t)=s(0)+\int_0^t \ln\left(u^2-3u+3\right)du$$

$$s(2)=8+\int_0^2 \ln\left(u^2-3u+3\right)du$$

=8.368 or 8.369

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln\left(u^2-3u+3\right)du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t)=s(0)+\int_0^t \ln\left(u^2-3u+3\right)du$$

$$s(2)=8+\int_0^2 \ln\left(u^2-3u+3\right)du$$

=8.368 or 8.369

8-2b

A particle moves on the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t)=12t^2-36t+15$. At $t=1$, the particle is at the origin.

24. Find the total distance traveled by the particle from $t=0$ to $t=2$.

Part D

1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\text{Total distance} = \int_0^{12} v\ (t)dt - \int_{\frac{1}{2}}^{2} v\ (t)dt = \left(x\left(\frac{1}{2}\right) - x(0)\right) - \left(x(2) - x\left(\frac{1}{2}\right)\right)$$

$$= 5\ 2 - (-1) - \left(-11 - \frac{5}{2}\right) = 17$$

0	1	2

The student response earns two of the following points:

1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\text{Total distance} = \int_0^{12} v\ (t)dt - \int_{\frac{1}{2}}^{2} v\ (t)dt = \left(x\left(\frac{1}{2}\right) - x(0)\right) - \left(x(2) - x\left(\frac{1}{2}\right)\right)$$

$$= 5\ 2 - (-1) - \left(-11 - \frac{5}{2}\right) = 17$$

8-2b

25. Find the position $x(t)$ of the particle at any time $t \geq 0$.

Part A

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$x(t) = 4t^3 - 18t^2 + 15t + C$
$0 = x(1) = 4 - 18 + 15 + C$
Therefore $C = -1$
$x(t) = 4t^3 - 18t^2 + 15t - 1$

0	1	2	3 ✓

The student response earns three of the following points:

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$x(t) = 4t^3 - 18t^2 + 15t + C$
$0 = x(1) = 4 - 18 + 15 + C$
Therefore $C = -1$
$x(t) = 4t^3 - 18t^2 + 15t - 1$

26. A particle travels along a straight line with a velocity of $v(t) = 3e^{\left(-\frac{t}{2}\right)} \sin(2t)$ meters per second. What is the total distance, in meters, traveled by the particle during the time interval $0 \leq t \leq 2$ seconds?

(A) 0.835
(B) 1.850
(C) 2.055
(D) 2.261 ✓
(E) 7.025

8-2b

27. A particle moves along the x-axis so that at any time $t \geq 0$ its velocity is given by $v(t) = \ln(t + 1) - 2t + 1$. The total distance traveled by the particle from $t = 0$ to $t = 2$ is

(A) 0.667

(B) 0.704

(C) 1.540 ✓

(D) 2.667

(E) 2.901

8-2b

A particle, initially at rest, moves along the x-axis so that its acceleration at any time $t \geq 0$ is given by $a(t) = 12t^2 - 4$. The position of the particle when $t=1$ is $x(1)=3$.

28. Find the total distance traveled by the particle from $t=0$ to $t=2$.

Part C

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	

Total Distance $\boxed{10}$

0	1	2

The student response earns two of the following points:

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

8-2b

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	
Total Distance		$\boxed{10}$

29. Write an expression for the position $x(t)$ of the particle at any time $t \geq 0$.

Part B

2 point is earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(1) = 1^4 - 2.1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

0	1	2	3

The student response earns three of the following points:

2 point is earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(1) = 1^4 - 2.1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

8-2b

30. Find the values of t for which the particle is at rest.

Part A

2 point is earned for $v\ (\ t\ )$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v\left(t\right)=0$

1 point is earned for finding two non-negative roots

$v\left(t\right)=4t^3-4t$
$v\left(t\right)=4t^3-4t=0$
$=4t\left(t^2-1\right)=0$
Therefore $t=0,t=1$

0	1	2	3	4

The student response earns four of the following points:

2 point is earned for $v\ (\ t\ )$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v\left(t\right)=0$

1 point is earned for finding two non-negative roots

$v\left(t\right)=4t^3-4t$
$v\left(t\right)=4t^3-4t=0$
$=4t\left(t^2-1\right)=0$
Therefore $t=0,t=1$

8-2b

A particle moves along the x-axis so that its velocity at time t is given by

$$v(t) = -(t+1)\sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.

31. Find the total distance traveled by the particle from time $t = 0$ until $t = 3$.

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

Distance $= \int_0^3 |v(t)| \mathrm{dt} = 4.333$ or 4.334

0	1	2	3

The student response earns three of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

Distance $= \int_0^3 |v(t)| \mathrm{dt} = 4.333$ or 4.334

8-2b

32. During the time interval $0 \le t \le 3$, what is the greatest distance between the particle and the origin? Show the work that leads to your answer.

Part D

1 point is earned for the ± distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t)\mathrm{dt} = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t)\mathrm{dt} = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

0	1	2

The student response earns two of the following points:

1 point is earned for the ± distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t)\mathrm{dt} = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t)\mathrm{dt} = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

8-2b

33. A particle moves along the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$. Find the position $x(t)$ of the particle at any time t if $x(0)=0$.

Part C

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$x(t) = \int v(t)dt$$
$$= \int [1 - sin(2\pi t)]\, dt$$
$$= t + \frac{1}{2\pi} cos(2\pi t) + C$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi}\cos(2\pi t) - \frac{1}{2\pi}$$

$$1 - \sin(2\pi t) = 0$$

$$t = \frac{1}{4} \text{ or } t = \frac{5}{4}$$

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

8-2b

$$x(t) = \int v(t)dt$$
$$= \int [\,1 - sin\,(2\pi t)]\,dt$$
$$= t + \frac{1}{2\pi}\,cos\,(2\pi t) + C$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi}\cos(2\pi t) - \frac{1}{2\pi}$$

$$1 - \sin(2\pi t) = 0$$

$$t = \frac{1}{4} \text{ or } t = \frac{5}{4}$$

8-2b

A particle moves along the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.

34. Write a polynomial expression for the position of the particle at any time $t \geq 0$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

< -1 > for error(s) in $t^3 - t^2 - t$

< -1 > for no constant of integration

$$x(t) = \int v(t)\mathrm{dt} = \int (3t^2 - 2t - 1)\,\mathrm{dt}$$
$$= t^3 - t^2 - t + C$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5;\ C = 3$$
$$x(t) = t^3 - t^2 - t + 3$$

0	1	2	3 ✓

The student response earns three of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

< -1 > for error(s) in $t^3 - t^2 - t$

< -1 > for no constant of integration

$$x(t) = \int v(t)\mathrm{dt} = \int (3t^2 - 2t - 1)\,\mathrm{dt}$$
$$= t^3 - t^2 - t + C$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5;\ C = 3$$
$$x(t) = t^3 - t^2 - t + 3$$

8-2b

35. Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

Part C

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\text{distance} = \int_0^3 |v(t)|\,\text{dt}$$
$$= \int_0^3 \left|3t^2 - 2t - 1\right|\text{dt} = 17$$

or

$$v(t) = 3t^2 - 2t - 1 = 0$$
$$t = -\frac{1}{3}, t = 1$$
$$x(0) = 1 - 1 - 1 + 3 = 2$$
$$x(3) = 27 - 9 - 3 + 3 = 18$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

distance = (3 - 2) + (18 - 2) = 17

0	1	2	3

The student response earns three of the following points:

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\text{distance} = \int_0^3 |v(t)|\,\text{dt}$$
$$= \int_0^3 \left|3t^2 - 2t - 1\right|\text{dt} = 17$$

or

$$v(t) = 3t^2 - 2t - 1 = 0$$
$$t = -\frac{1}{3}, t = 1$$
$$x(0) = 1 - 1 - 1 + 3 = 2$$
$$x(3) = 27 - 9 - 3 + 3 = 18$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

8-2b

distance = (3 - 2) + (18 - 2) = 17

A particle moves along the x-axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t)=4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0)=1$ and its position is $x(0)=0$.

36. Write an equation for the position $x(t)$ of the particle.

Part B

1 point is earned for integrand

1 point is earned for $x\left(t\right) = -\cos 2t + t + C$ or $x\left(t\right) = -\cos 2t + t$

1 point is earned for answer

$$x\left(t\right) = \int 2\sin 2t + 1dt$$
$$x\left(t\right) = -\cos 2t + t + C$$
$$x\left(0\right) = 0 \Rightarrow C = 1$$
$$x\left(t\right) = -\cos 2t + t + 1$$

0	1	2	3 ✓

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $x\left(t\right) = -\cos 2t + t + C$ or $x\left(t\right) = -\cos 2t + t$

1 point is earned for answer

$$x\left(t\right) = \int 2\sin 2t + 1dt$$
$$x\left(t\right) = -\cos 2t + t + C$$
$$x\left(0\right) = 0 \Rightarrow C = 1$$
$$x\left(t\right) = -\cos 2t + t + 1$$

8-2b

37. Write an equation for the velocity $v(t)$ of the particle.

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$v(t) = \int 4\cos 2t\, dt$$
$$v(t) = 2\sin 2t + C$$
$$v(0) = 1 \Rightarrow C = 1$$
$$v(t) = 2\sin 2t + 1$$

0	1	2	3

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$v(t) = \int 4\cos 2t\, dt$$
$$v(t) = 2\sin 2t + C$$
$$v(0) = 1 \Rightarrow C = 1$$
$$v(t) = 2\sin 2t + 1$$

8-2b

An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

38. What is the total distance traveled by the object over the time interval $0 \leq t \leq 4$?

Part C

1 point is earned for limits of 0 and 4 on an integral

of v(t) or $|v(t)|$

or uses x(0) and x(4) to compute distance

1 point is earned for handles change of direction at student's turning point answer

1 point is earned for the answer

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for limits of 0 and 4 on an integral

of v(t) or $|v(t)|$

or uses x(0) and x(4) to compute distance

1 point is earned for handles change of direction at student's turning point answer

1 point is earned for the answer

8-2b

39. What is the position of the object at time $t = 4$?

Part D

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x\left(t\right)=-\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right)+C$

1 point is earned for the answer

$$x\left(4\right)=x\left(0\right)+\int_0^4 v\left(t\right)dt=3.432$$

OR

$$x\left(t\right)=-\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right)+\frac{3}{\pi}+2$$

$$x\left(4\right)=2+\frac{9}{2\pi}=3.432$$

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x\left(t\right)=-\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right)+C$

1 point is earned for the answer

8-2b

$$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$$

OR

$$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$

$$x(4) = 2 + \frac{9}{2\pi} = 3.432$$

8-2b

A particle moves long the x-axis so that its velocity *v* at any time *t,* for $0 \leq t \leq 16$ is given by $v(t) = e^{2\sin t} - 1$. At time $t = 0$, the particle is at the origin.

40. Find the total distance traveled by the particle from $t = 0$ to $t = 4$.

Part C

1 point is earned for the limits of 0 and 4 on an integral of v(t) or $|v(t)|$

or uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

$$\int_0^4 |v(t)|dt = 10.542$$

or

$$v(t) = e^{2\sin t} - 1 = 0$$

$$t = 0, or, t = \pi$$

$$x(\pi) = \int_0^\pi v(t)dt = 10.10656$$

$$x(4) = \int_0^4 v(t)dt = 9.67066$$

$$|x(\pi) - x(0)| + |x(4) - x(\pi)|$$

$$= 10.542$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 4 on an integral of v(t) or $|v(t)|$

or uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

8-2b

$$\int_0^4 |v(t)|dt = 10.542$$

or

$$v(t) = e^{2\sin t} - 1 = 0$$
$$t = 0, or, t = \pi$$
$$x(\pi) = \int_0^{\pi} v(t)dt = 10.10656$$
$$x(4) = \int_0^4 v(t)dt = 9.67066$$
$$|x(\pi) - x(0)| + |x(4) - x(\pi)|$$
$$= 10.542$$

41. Is there any time t, $0 \le t \le 16$, at which the particle returns to the origin? Justify your answer.

Part D

1 point is earned for no such time

1 point is earned for the reason

There is no such time because

$$\int_0^T v(t)dt > 0 \text{ for all } T > 0.$$

0	1	2

The student response earns all of the following points:

1 point is earned for no such time

1 point is earned for the reason

There is no such time because

$$\int_0^T v(t)dt > 0 \text{ for all } T > 0.$$

8-2b

42. A particle moves along the x-axis. The velocity of the particle at time t is $6t - t^2$. What is the total distance traveled by the particle from time $t = 0$ to $t = 3$?

(A) 3

(B) 6

(C) 9

(D) 18 ✓

(E) 27

43. A particle with velocity at any time t given by $v(t) = e^t$ moves in a straight line. How far does the particle move from $t = 0$ to $t = 2$?

(A) $e^2 - 1$ ✓

(B) $e - 1$

(C) $2e$

(D) e^2

(E) $\frac{e^3}{3}$

8-2b

44. A particle moves along the y-axis with velocity given by $v(t) = t\sin(t^2)$ for $t \geq 0$.

Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part D

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or

or $|v(t)|$

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

$$dis\tan ce = \int_0^2 |v(t)|dt = 1.173$$

$$or$$

$$v(t) = t\sin t^2 = 0$$

$$t = 0, or, t = \sqrt{\pi} \approx 1.772$$

$$y(0) = 3; y\left(\sqrt{\pi}\right) = 4; y(2) = 3.826, or, 3.827$$

$$\left[y\left(\sqrt{\pi}\right) - y(0)\right] + \left[y\left(\sqrt{\pi}\right) - y(2)\right]$$

$$= 1.173, or, 1.174$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or

or $|v(t)|$

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

8-2b

$$dis\tan ce = \int_0^2 |v(t)|dt = 1.173$$

or

$$v(t) = t\sin t^2 = 0$$
$$t = 0, or, t = \sqrt{\pi} \approx 1.772$$
$$y(0) = 3; y\left(\sqrt{\pi}\right) = 4; y(2) = 3.826, or, 3.827$$
$$\left[y\left(\sqrt{\pi}\right) - y(0)\right] + \left[y\left(\sqrt{\pi}\right) - y(2)\right]$$
$$= 1.173, or, 1.174$$

45. At t = 0 a particle starts at rest and moves along a line in such a way that at time t its acceleration is $24t^2$ feet per second per second. Through how many feet does the particle move during the first 2 seconds?

(A) 32 ✓

(B) 48

(C) 64

(D) 96

(E) 192

46. The acceleration α of a body moving in a straight line is given in terms of time t by α=8−6t. If the velocity of the body is 25 at t = 1 and if $s(t)$ is the distance of the body from the origin at time t, what is $s(4) - s(2)$?

(A) 20

(B) 24

(C) 28

(D) 32 ✓

(E) 42

47. If the velocity of a particle moving along the x-axis is $v(t) = 2t - 4$ and if at t=0 its position is 4, then at any time t its position $x(t)$ is

(A) t^2-4t

(B) t^2-4t-4

(C) t^2-4t+4 ✓

(D) 2t^2-4t

(E) 2t^2-4t+4

8-2b

48. The acceleration of a particle moving along the x-axis at time t is given by $a(t) = 6t - 2$. If the velocity is 25 when t=3 and the position is 10 when t=1, then the position $x(t)$=

(A) $9t^2 + 1$

(B) $3t^2 - 2t + 4$

(C) $t^3 - t^2 + 4t + 6$ ✓

(D) $t^3 - t^2 + 9t - 20$

(E) $36t^3 - 4t^2 - 77t + 55$

Answer C

49. A particle moves along the x-axis so that at any time $t \geq 0$ the acceleration of the particle is $a(t) = e^{-2t}$. If at t=0 the velocity of the particle is $\frac{5}{2}$ and its position is $\frac{17}{4}$, then its position at any time t>0 is $x(t)$=

(A) $-\frac{e^{-2t}}{2} + 3$

(B) $\frac{e^{-2t}}{4} + 4$

(C) $4e^{-2t} + \frac{9}{2}t + \frac{1}{4}$

(D) $\frac{e^{-2t}}{2} + 3t + \frac{15}{4}$

(E) $\frac{e^{-2t}}{4} + 3t + 4$ ✓

50. An object traveling in a straight line has position $x(t)$ at time t.If the initial position is $x(0) = 2$ and the velocity of the object is $v(t) = \sqrt[3]{1 + t^2}$, what is the position of the object at time $t = 3$?

(A) 0.431

(B) 2.154

(C) 4.512

(D) 6.512 ✓

(E) 17.408

8-2b

51. A particle moves on the x-axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$. Find the total distance traveled by the particle from t=0 to t=5 .

Part C

1 point is earned for correctly solving $x''(t) = 0$ or otherwise identifying turning point

$x'(t) = v(t) = 2e^{-t}(1 - t) = 0$
$t = 1$

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and x(turning point)

$x(0) = 0$
$x(1) = \frac{2}{e} \approx 0.736$
$x(5) = \frac{10}{e^5} \approx 0.067$

1 point is earned for the correct answer

Distance $= \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$
≈ 1.404

Note: Max 1/3 if no turning point or turning point ignored

0	1	2	3

The student response earns three of the following points:

1 point is earned for correctly solving $x''(t) = 0$ or otherwise identifying turning point

$x'(t) = v(t) = 2e^{-t}(1 - t) = 0$
$t = 1$

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and x(turning point)

$x(0) = 0$
$x(1) = \frac{2}{e} \approx 0.736$
$x(5) = \frac{10}{e^5} \approx 0.067$

1 point is earned for the correct answer

8-2b

Distance $= \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$
≈ 1.404

Note: Max 1/3 if no turning point or turning point ignored

52. A particle moves along the x-axis so that at any time $t \geq 0$, its velocity is given by $v(t) = \cos(2 - t^2)$. The position of the particle is 3 at time $t = 0$. What is the position of the particle when its velocity is first equal to 0?

(A) 0.411
(B) 1.310
(C) 2.816 ✓
(D) 3.091
(E) 3.411

53. A particle moves along the x-axis with velocity given by $v(t)= 3t^2 + 6t$ for time $t \geq 0$. If the particle is at position $x = 2$ at time $t = 0$, what is the position of the particle at time $t = 1$?

(A) 4
(B) 6 ✓
(C) 9
(D) 11
(E) 12

8-2b

For $0 \le t \le 12$, a particle moves along the x-axis. The velocity of the particle at time t is given by $v(t) = \cos(\pi/6t)$. The particle is at position $x = -2$ at time $t = 0$.

54. Write, but do not evaluate, an integral expression that gives the total distance traveled by the particle from time $t = 0$ to time $t = 6$.

Part B

1 point is earned for answer

$$\int_0^6 \left| v(t) \right| dt$$

0	1

The student response earns all of the following points:

1 point is earned for answer

$$\int_0^6 \left| v(t) \right| dt$$

8-2b

55. Find the position of the particle at time $t = 4$.

Part D

The student response earns none of the following points:

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$x(4) = -2 + \int_0^4 \cos(\tfrac{\pi}{6}t)dt$$
$$= -2 + \left[\tfrac{6}{\pi}\sin\left(\tfrac{\pi}{6}t\right)\right]_0^4$$
$$= -2 + \tfrac{6}{\pi}\left[\sin\left(\tfrac{2\pi}{3}\right) - 0\right]$$
$$= -2 + \tfrac{6}{\pi} \cdot \tfrac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$x(4) = -2 + \int_0^4 \cos(\tfrac{\pi}{6}t)dt$$
$$= -2 + \left[\tfrac{6}{\pi}\sin\left(\tfrac{\pi}{6}t\right)\right]_0^4$$
$$= -2 + \tfrac{6}{\pi}\left[\sin\left(\tfrac{2\pi}{3}\right) - 0\right]$$
$$= -2 + \tfrac{6}{\pi} \cdot \tfrac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}$$

8-2b

56. A particle moves along a straight line. For $0 \leq t \leq 5$, the velocity of the particle is given by $v(t) = -2 + (t^2 + 3t)^{6/5} - t^3$, and the position of the particle is given by $s(t)$. It is known that $s(0) = 10$.

Write an expression involving an integral that gives the position $s(t)$. Use this expression to find the position of the particle at time $t = 5$.

Part B

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x)dx$$

$$s(5) = 10 + \int_0^5 v(x)dx = -9.207$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x)dx$$

$$s(5) = 10 + \int_0^5 v(x)dx = -9.207$$

57. A particle moves along a line so that its acceleration for $t \geq 0$ is given by $a(t) = \frac{t+3}{\sqrt{t^3+1}}$. If the particle's velocity at $t = 0$ is 5, what is the velocity of the particle at $t = 3$?

(A) 0.713

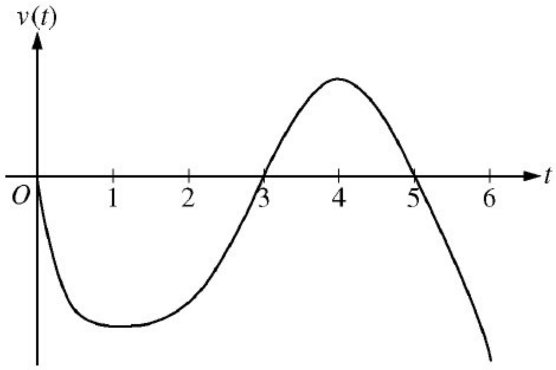
(B) 1.134

(C) 6.134

(D) 6.710

(E) 11.710 ✓

8-2b



Graph of v

A particle moves along the x-axis so that its velocity at time t, for $0 \le t \le 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t-axis and the graph of v on the intervals [0, 3], [3, 5], and [5, 6] are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

58. For how many values of t, where $0 \le t \le 6$, is the particle at $x = -8$? Explain your reasoning.

Part B

The response can earn up to 3 points:

1 point: For the positions at $t = 3$, $t = 5$, and $t = 6$

1 point: For the description of motion

1 point: For the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

0	1	2	3

The response earns all three of the following points:

1 point: For the positions at $t = 3$, $t = 5$, and $t = 6$

1 point: For the description of motion

1 point: For the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

8-2b

59. For $0 \leq t \leq 6$, find both the time and the position of the particle is farthest to the left. Justify your answer.

Part A

The response can earn up to 3 points:

1 point: For identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 > t > 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point: For consideration of $\int_0^6 v(t)\,dt$

$$x(3) = -2 + \int_0^3 v(t)\,dt = -2 - 8 = -10x(6) = -2 + \int_0^6 v(t)\,dt = -2 - 8 + 3 - 2 = -9$$

1 point: For the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

0	1	2	3

The response earns all three of the following points:

1 point: For identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 > t > 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point: For consideration of $\int_0^6 v(t)\,dt$

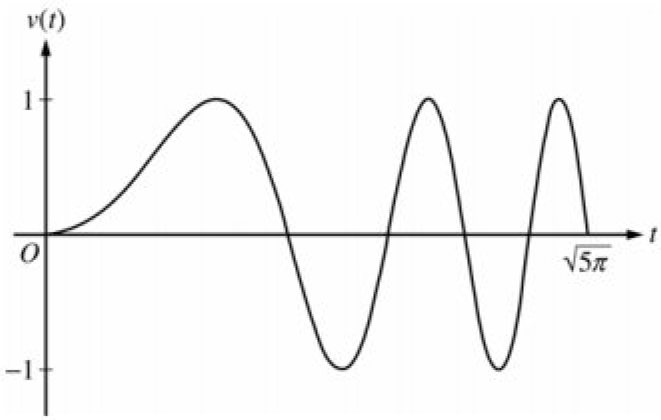
$$x(3) = -2 + \int_0^3 v(t)\,dt = -2 - 8 = -10x(6) = -2 + \int_0^6 v(t)\,dt = -2 - 8 + 3 - 2 = -9$$

1 point: For the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

8-2b

60.



A particle moves along the x-axis so that its velocity v at time $t \geq 0$ is given by $v(t) = \sin(t^2)$. The graph of v is shown above for $0 \leq t \leq \sqrt{5\pi}$. The position of the particle at time t is $x(t)$ and its position at time $t = 0$ is $x(0) = 5$.

Find the total distance traveled by the particle from time t=0 to t=3.

Part B

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^9 \left| v(t) \right| dt = 1.702$$

OR

$$For\ 0 < t < 3,\ v(t) = 0\ when\ t = \sqrt{\pi} = 1.77245\ and$$
$$t = \sqrt{2\pi} = 2.50663$$
$$x(0) = 5$$
$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t)dt = 5.89483$$
$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t)dt = 5.43041$$
$$x(3) = 5 + \int_0^{3} v(t)dt = 5.77356$$
$$\left| x(\sqrt{\pi}) - x(0) \right| + \left| x(\sqrt{2\pi}) - x(\sqrt{\pi}) \right| + \left| x(3) - x(\sqrt{2\pi}) \right| = 1.702$$

0	1	2

8-2b

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

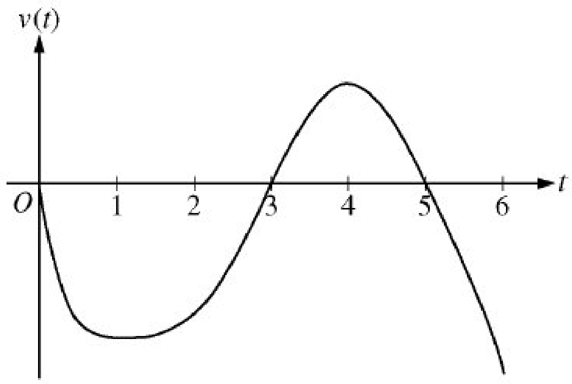
Distance$= \displaystyle\int_0^9 \left|v(t)\right| dt = 1.702$

OR

$$For\, 0 < t < 3,\ v(t) = 0\, when\, t = \sqrt{\pi} = 1.77245\, and$$
$$t = \sqrt{2\pi} = 2.50663$$
$$x(0) = 5$$
$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t)dt = 5.89483$$
$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t)dt = 5.43041$$
$$x(3) = 5 + \int_0^{3} v(t)dt = 5.77356$$
$$\left|x(\sqrt{\pi}) - x(0)\right| + \left|x(\sqrt{2\pi}) - x(\sqrt{\pi})\right| + \left|x(3) - x(\sqrt{2\pi})\right| = 1.702$$

8-2b

61.



Graph of v

A particle moves along the x-axis so that its velocity at time t, for $0 \le t \le 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t-axis and the graph of v on the intervals [0,3], [3,5], and [5,6] are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

For how many values of t, where $0 \le t \le 6$, is the particle at x = −8 ? Explain your reasoning.

Part B

3 points can be earned.

1 point for positions at $t = 3$, $t = 5$ and $t = 6$.

1 point for description of motion.

1 point for conclusion.

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

0	1	2	3

3 points can be earned.

1 point for positions at $t = 3$, $t = 5$ and $t = 6$.

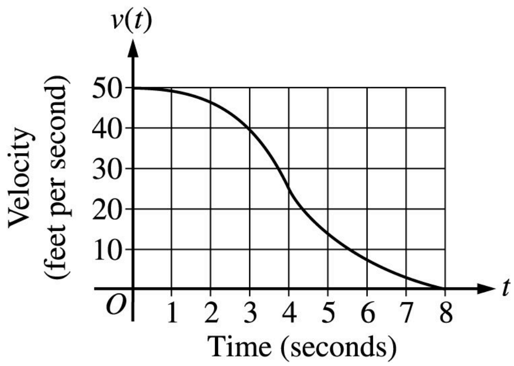
1 point for description of motion.

1 point for conclusion.

8-2b

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

62.



The graph above gives the velocity, in in ft/sec, of a car for $0 \leq t \leq 8$, where t is the time in seconds. Of the following, which is the best estimate of the distance traveled by the car from $t = 0$ until the car comes to a complete stop?

(A) 21 ft

(B) 26 ft

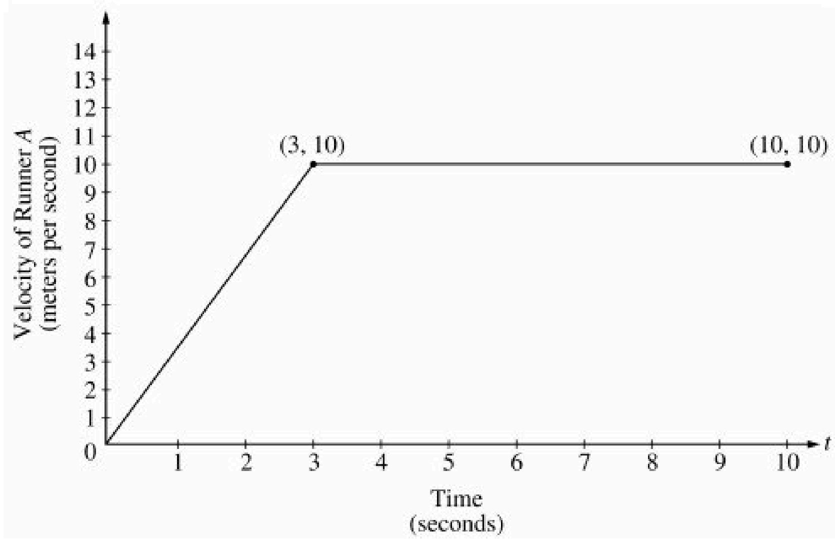
(C) 180 ft

(D) 210 ft ✓

(E) 260 ft

8-2b

63.



Two runners, A and B, run on a straight racetrack for $0 \leq t \leq 10$ seconds. The graph above, which consists of two line segments. shows the velocity, in meters per second, of Runner A. The velocity, in meters per second, of Runner B is given by the function v defined by $v(t) = \frac{24t}{2t+3}$.

Find the total distance run by Runner A and the total distance run by Runner B over the time interval $0 \leq t \leq 10$ seconds. Indicate units of measure.

Part C

2 points are earned for distance for Runner A

1 : method

1 : answer

2 points are earned for distance for Runner B

1 : integral

1 : answer

Runner A: distance $= \frac{1}{2}(3)(10) + 7(10) = 85 meters$

Runner B: distance $= \int_0^{10} \frac{24t}{2t+3} dt = 83.336 meters$

1 point is earned for the units

(units) meters/sec in part (a), meter/sec2, meters/sec2 in part (b), and meters in part (c), or equivalent.

8-2b

0	1	2	3	4	5

The student response earns all of the following points:

2 points are earned for distance for Runner *A*

1 : method

1 : answer

2 points are earned for distance for Runner *B*

1 : integral

1 : answer

Runner *A*: distance $= \frac{1}{2}(3)\,(10) + 7\,(10) = 85 meters$

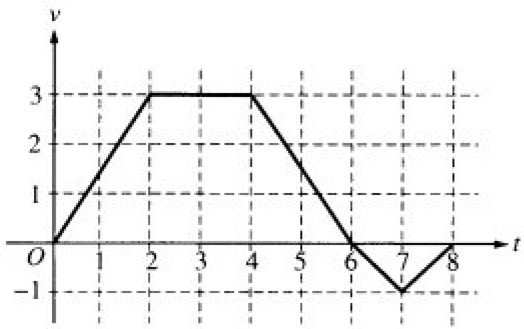
Runner *B*: distance $= \int_0^{10} \frac{24t}{2t+3} dt = 83.336 meters$

1 point is earned for the units

(units) meters/sec in part (a), meter/sec2, meters/sec2 in part (b), and meters in part (c), or equivalent.

8-2b

64.



A bug begins to crawl up a vertical wire at time $t = 0$. The velocity v of the bug at time t, $0 \le t \le 8$, is given by the function whose graph is shown above.

What is the total distance the bug traveled from $t = 0$ to $t = 8$

(A) 14

(B) 13

(C) 11

(D) 8

(E) 6

8-2b

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

65. Using correct units, explain the meaning of $\int_{10}^{70} v(t)dt$ in terms of the rocket's flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t)dt$.

Part B

1 point is earned for the explanation

1 point is earned for uses v(20), v(40), v(60)

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

20[v(20) + v(40) + v(60)]

= 20[22 + 35 + 44] = 2020 ft

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for uses v(20), v(40), v(60)

1 point is earned for the value

8-2b

Since the velocity is positive, $\int_{10}^{70} v(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

20[v(20) + v(40) + v(60)]

= 20[22 + 35 + 44] = 2020 ft

8-2b

66. Rocket B is launched upward with an acceleration of $a(t) = \frac{3}{\sqrt{t+1}}$ feet per second per second. At time $t =$ 0 seconds, the initial height of the rocket is 0 feet, and the initial velocity is 2 feet per second. Which of the two rockets is traveling faster at time $t = 80$ seconds? Explain your answer.

Part C

1 point is earned for:$6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $vB(80)$, compares v(80), and draws a conclusion

Let $vB(t)$ be the velocity of rocket B at time t.

$$v_B(t) \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$
$$2 = v_B(0) = 6 + C$$
$$v_B(t) = 6\sqrt{t+1} - 4$$
$$v_B(80) = 50 > 49 = v_B(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for:$6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $vB(80)$, compares v(80), and draws a conclusion

Let $vB(t)$ be the velocity of rocket B at time t.

8-2b

$v_B(t) \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$

$2 = v_B(0) = 6 + C$

$v_B(t) = 6\sqrt{t+1} - 4$

$v_B(80) = 50 > 49 = v_B(80)$

Rocket B is traveling faster at time $t = 80$ seconds.

Units

1 point is earned for units in (a) and (b)

Units of ft/sec2 in (a) and ft in (b)

0	1

The student response earns all of the following points:

1 point is earned for units in (a) and (b)

Units of ft/sec2 in (a) and ft in (b)

8-2b

67.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	–220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v. Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

Using correct units, explain the meaning of the definite integral $\int_{0}^{40} |v(t)|dt$ in the context of the problem. Approximate the value of $\int_{0}^{40} |v(t)|dt$ using a right Riemann sum with the four subintervals indicated in the table.

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_{40}^{0} |v(t)|dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$\int_{40}^{0} |v(t)|dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$

$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$

$= 2400 + 1920 + 880 + 2400$

$= 7600$ meters

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

8-2b

$\int_{40}^{0} |v(t)|dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$\int_{40}^{0} |v(t)|dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$

$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$

$= 2400 + 1920 + 880 + 2400$

$= 7600$ meters

8-2b

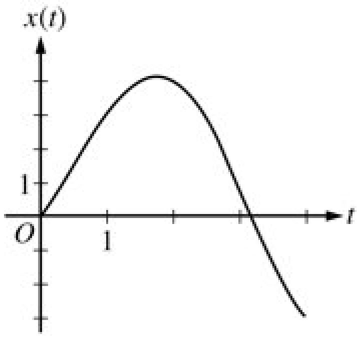
68.

t	0	1	2	3	4
$v(t)$	-1	2	3	0	-4

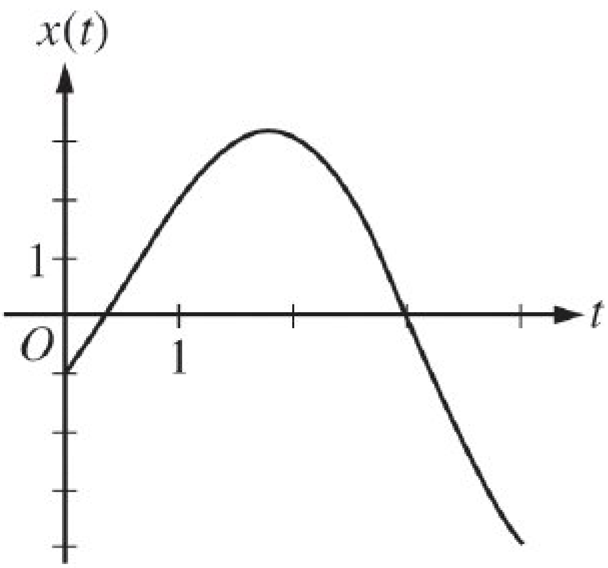
The table gives selected values of the velocity, $v(t)$, of a particle moving along the x-axis. At time $t = 0$, the particle is at the origin. Which of the following could be the graph of the position, $x(t)$, of the particle for $0 \le t \le 4$?

8-2b

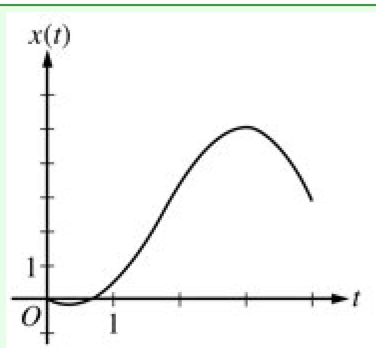
(A)



(B)

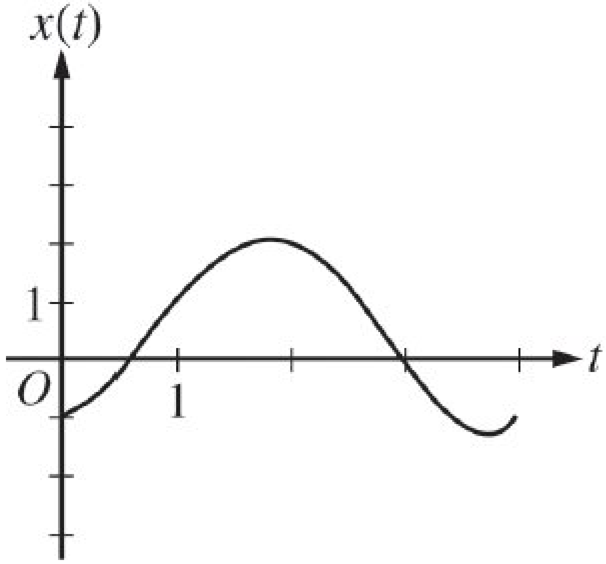


(C)

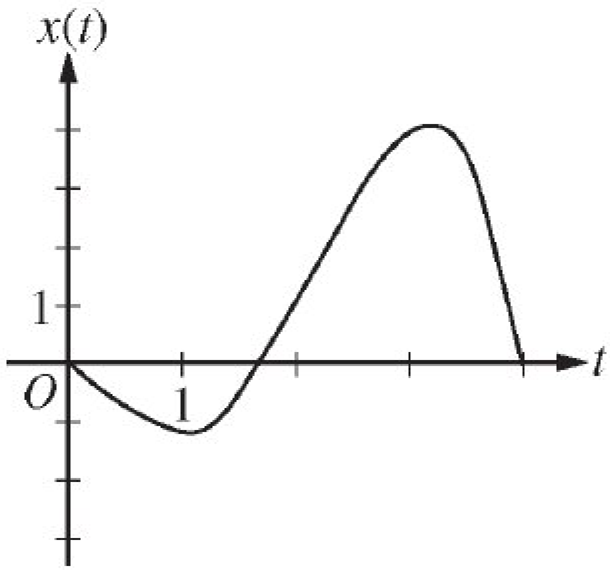


8-2b

(D)



(E)



8-2b

69.

t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/minute)	0	100	40	−120	−150

Train A runs back and forth on an east-west section of railroad track. Train A's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

At time $t = 2$, train A's position is 300 meters east of the Origin Station, and the train is moving to the east. Write an expression involving an integral that gives the position of train A, in meters from the Origin Station, at time $t = 12$. Use a trapezoidal sum with three subintervals indicated by the table to approximate the position of the train at time $t = 12$.

Part C

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s^A(12) = s^A(2) + \int_2^{12} V_A(t) = 300 + \int_2^{12} V_A(t)dt$$

$$\int_2^{12} V_A(t)dt \approx 3.\frac{100+40}{2} + 3.\frac{40-120}{2} + 4.\frac{-120-150}{2}$$

$S_A(12) \approx 300 - 450 = -150$

The position of Train A at time t=12 minutes is approximately 150 meters west of Origin Station.

0	1	2	3

The student response earns three of the following points:

1 point is earned for the position expression

1 point is earned for the trapezoidal sum

1 point is earned for position at time $t = 12$

$$s^A(12) = s^A(2) + \int_2^{12} V_A(t) = 300 + \int_2^{12} V_A(t)dt$$

8-2b

$$\int_{2}^{12} V_A(t)dt \approx 3.\frac{100+40}{2} + 3.\frac{40-120}{2} + 4.\frac{-120-150}{2}$$

= -450

S_A (12) ≈ 300 - 450 = -150

The position of Train *A* at time *t* = 12 minutes is approximately 150 meters west of Origin Station.

70. A particle travels in a straight line with a constant acceleration of 3 meters per second per second. If the velocity of the particle is 10 meters per second at time 2 seconds, how far does the particle travel during the time interval when its velocity increases from 4 meters per second to 10 meters per second?

(A) 20 m

(B) 14 m

(C) 7 m

(D) 6 m

(E) 3 m

8-2b

71.

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	–10	–8	–4	7

The velocity of a particle moving along the x-axis is modeled by a differentiable function v, where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position $x = 7$ meters when $t = 0$ seconds.

(a) Estimate the acceleration of the particle at $t = 36$ seconds. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{20}^{40} v(t)\,dt$ in the context of this problem. Use a trapezoidal sum with the three subintervals indicated by the data in the table to approximate $\int_{20}^{40} v(t)\,dt$.

(c) For $0 \leq t \leq 40$, must the particle change direction in any of the subintervals indicated by the data in the table? If so, identify the subintervals and explain your reasoning. If not, explain why not.

(d) Suppose that the acceleration of the particle is positive for $0 < t < 8$ seconds. Explain why the position of the particle at $t = 8$ seconds must be greater than $x = 30$ meters.

Part A

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

1 point is earned for units in (a) and (b)

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

1 point is earned for units in (a) and (b)

8-2b

Part B

1 point is earned for meaning of $\int_{20}^{40} v(t)dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t)dt$ is the particle's change in position in meters from time $t = 20$ seconds to time $t = 40$ seconds.

$$\int_{20}^{40} v(t)dt \approx \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$
$$= -75 \text{ meters}$$

1 point is earned for units in (a) and (b)

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for meaning of $\int_{20}^{40} v(t)dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t)dt$ is the particle's change in position in meters from time $t = 20$ seconds to time $t = 40$ seconds.

$$\int_{20}^{40} v(t)dt \approx \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$
$$= -75 \text{ meters}$$

1 point is earned for units in (a) and (b)

Part C

1 point is earned for the answer

1 point is earned for the explanation

$v(8) > 0$ and $v(20) < 0$

$v(32) < 0$ and $v(40) > 0$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.

8-2b

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

$v(8) > 0$ and $v(20) < 0$

$v(32) < 0$ and $v(40) > 0$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.

Part D

1 point is earned for $v'(t) = a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t) = a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

Therefore, $x(8) = x(0) + \int_0^8 v(t)dt \geq 7 + 8 \cdot 3 > 30.$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $v'(t) = a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t) = a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

Therefore, $x(8) = x(0) + \int_0^8 v(t)dt \geq 7 + 8 \cdot 3 > 30.$

8-2b

72. A particle moves along the x-axis so that at any time $t > 0$ its velocity is given by $v(t) = t \ln t - t$. At time $t = 1$, the position of the particle is $x(1) = 6$.

(a) Write an expression for the acceleration of the particle.

(b) For what values of t is the particle moving to the right?

(c) What is the minimum velocity of the particle? Show the analysis that leads to your conclusion.

(d) Write an expression of the position $x(t)$ of the particle.

Part A

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

0	1 ✓

The student earns all of the following points:

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

Part B

1 point is earned for correctly setting $t \ln t - t > 0$

$v(t) = t \ln t - t > 0$

$= t(\ln t - 1) > 0$

1 point is earned for the correct answer

$t > e$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for correctly setting $t \ln t - t > 0$

$v(t) = t \ln t - t > 0$

8-2b

$= t\,(\ln t - 1) > 0$

1 point is earned for the correct answer

$t > e$

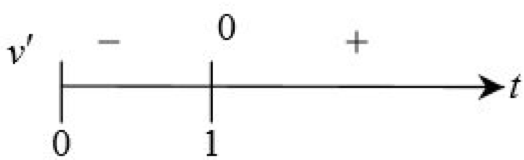
Part C

1 point is earned for correctly solving $v'(t) = 0$

$v'(t) = \ln t = 0$

$t = 1$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is $v(1) = -1$

Max 1/3 if student's v' does not involve ln

0	1	2	3

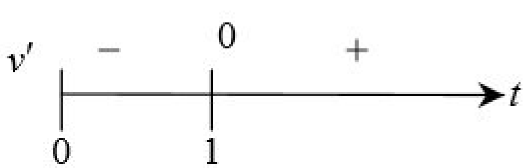
The student response earns all of the following points:

1 point is earned for correctly solving $v'(t) = 0$

$v'(t) = \ln t = 0$

$t = 1$

1 point is earned for the correct analysis



1 point is earned for the correct answer

8-2b

minimum velocity is $v(1) = -1$

Max 1/3 if student's v' does not involve ln

Part D

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

<-1>: any other error

$$\int t \ln t - t \, dt$$
$$= \left(\frac{1}{2}t^2 \ln t - \frac{1}{4}t^2\right) - \frac{1}{2}t^2 + C$$
$$= \frac{1}{2}t^2 \ln t - \frac{3}{4}t^2 + C;$$
$$6 = x(1) = 0 - \frac{3}{4} + C$$

1 point is earned for correctly solving for C

$$C = \frac{27}{4};$$
$$x(t) = \frac{1}{2}t^2 \ln t - \frac{3}{4}t^2 + \frac{27}{4}$$

0/3 if not antidifferentiating $t \ln t - t$

0	1	2	3 ✓

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

<-1>: any other error

8-2b

$$\int t \ln t - t\, dt$$
$$= \left(\tfrac{1}{2}t^2 \ln t - \tfrac{1}{4}t^2\right) - \tfrac{1}{2}t^2 + C$$
$$= \tfrac{1}{2}t^2 \ln t - \tfrac{3}{4}t^2 + C;$$
$$6 = x(1) = 0 - \tfrac{3}{4} + C$$

1 point is earned for correctly solving for *C*

$$C = \frac{27}{4};$$
$$x(t) = \tfrac{1}{2}t^2 \ln t - \tfrac{3}{4}t^2 + \frac{27}{4}$$

0/3 if not antidifferentiating *t* ln *t* - *t*

8-2b

73. Two particles move along the x-axis. For $0 \le t \le 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.

(a) For $0 \le t \le 6$, find all times t during which particle R is moving to the right.

(b) For $0 \le t \le 6$, find all times t during which the two particles travel in opposite directions.

(c) Find the acceleration of particle P at time $t = 3$. Is particle P speeding up, slowing down, or doing neither at time $t = 3$? Explain your reasoning.

(d) Write, but do not evaluate, an expression for the average distance between the two particles on the interval $1 \le t \le 3$.

Part A

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$$

$r'(t) = 0$ when $t = 1$ and $t = 3$

$r'(t) > 0$ for $0 < t < 1$ and $3 < t < 6$

$r'(t) < 0$ for $1 < t < 3$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.

0	1	2

The student response earns all of the following points:

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$$

$r'(t) = 0$ when $t = 1$ and $t = 3$

$r'(t) > 0$ for $0 < t < 1$ and $3 < t < 6$

$r'(t) < 0$ for $1 < t < 3$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.

8-2b

Part B

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4}\sin\left(\frac{\pi}{4}t\right)$$

$p'(t) = 0$ when $t = 0$ and $t = 4$

$p'(t) < 0$ for $0 < t < 4$

$p'(t) > 0$ for $4 < t < 6$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4}\sin\left(\frac{\pi}{4}t\right)$$

$p'(t) = 0$ when $t = 0$ and $t = 4$

$p'(t) < 0$ for $0 < t < 4$

$p'(t) > 0$ for $4 < t < 6$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.

Part C

1 point is earned for $p''(3)$

8-2b

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4} t\right)$$
$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$p'(3) < 0$

Therefore particle P is slowing down at time $t = 3$.

0	1	2

The student response earns all of the following points:

1 point is earned for $p''(3)$

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4} t\right)$$
$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$p'(3) < 0$

Therefore particle P is slowing down at time $t = 3$.

Part D

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integrand

8-2b

1 point is earned for the limits and constant

$$\frac{1}{2}\int_{1}^{3}|p(t)-r(t)|dt$$

8-2b

74. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x-axis such that, for time $t > 0$, its position is given by $x_P(t) = 6 - 4e^{-t}$.

Particle Q moves along the y-axis such that, for time $t > 0$, its velocity is given by $v_Q(t) = \frac{1}{t^2}$. At time $t = 1$, the position of particle Q is $y_Q(1) = 2$.

(a) Find $v_P(t)$, the velocity of particle P at time t.

(b) Find $a_Q(t)$, the acceleration of particle Q at time t. Find all times t, for $t > 0$, when the speed of particle Q is decreasing. Justify your answer.

(c) Find $y_Q(t)$, the position of particle Q at time t.

(d) As $t \to \infty$, which particle will eventually be farther from the origin? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that equates $x_P(t)$ with $v_P(t)$ does not earn the point.

An unlabeled response earns the point.

✓

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$v_P(t) = x_P'(t) = 4e^{-t}$

Part B

8-2b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earning the first point is not necessary for a response to be eligible to earn the second or third points; however, the response must present an expression for $a_Q(t)$ to be eligible for third point.

A response earns the second point with either of the following statements: "$v_Q(t)$ and $a_Q(t)$ have opposite signs" or "$v_Q(t)$ and $a_Q(t)$ have the same sign." This statement, however, must be consistent with $v_Q(t)$ and the presented expression for $a_Q(t)$.

A response must earn the second point to be eligible for the third point. The answer must be consistent with the presented justification. Furthermore, responses for which $a_Q(t) > 0$ for $t > 0$ must conclude that there is no time at which the speed of the particle is decreasing.

A response that indicates $v_Q\ (t) < 0$ does not earn the third point, even if the answer and justification are consistent with a reported sign of $a_Q(t)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] $a_Q(t)$
- [ ] Considers signs of $a_Q(t)$ and $v_Q(t)$
- [ ] Answer with justification

Solution:

$$a_Q(t) = v_Q{}'(t) = \frac{-2}{t^3}$$

For $t > 0$, $a_Q(t) < 0$ and $v_Q(t) > 0$.

Because the velocity and acceleration have opposite signs, the speed of particle Q is decreasing for all $t > 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $\int_1^t \frac{1}{t^2}\,dt$ (using the same variable as a limit and integrand function) does not earn the first point unless it is followed by an attempt at integration.

A response that presents either $\int \frac{1}{t^2}\,dt$ or $-\frac{1}{t}$ (with no integral) earns the first point. If the response continues and presents $2 = -1 + C$, then the response earns the second point.

8-2b

A response that presents only $y_Q(t) = -\frac{1}{t} + 3$ will earn all 3 points. Note that the right side of this equation suffices to earn all points. A response of $y_Q(t) = -\frac{1}{t} + C$, where $C \neq 3$, (with no additional supporting work) earns only the first point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Integral
- [ ] Uses initial condition
- [ ] Answer

Solution:

$$y_Q(t) = y_Q(1) + \int_1^t \frac{1}{s^2}\, ds$$

$$= 2 - \left(\frac{1}{s}\Big|_1^t\right) = 2 - \frac{1}{t} + 1 = 3 - \frac{1}{t}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with an incorrect $y_Q(t)$ from part (c) is eligible for both points in part (d) provided $y_Q(t)$ is a non-constant function. The second point is earned for a consistent answer with reason, and limits correct for particle P and the presented $y_Q(t)$.

Responses that present statements such as "$6 - 4e^{-t}$ approaches 6" or "Q goes to 3" earn the first point and are eligible for the second point.

A response that treats ∞ as an input for $x_P(t)$ or $y_Q(t)$, such as "$6 - 4e^{-\infty}$" or "$3 - \frac{1}{\infty}$" is not eligible for the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] One correct limit
- [ ] Answer with reason

8-2b

Solution:

For particle P, $\lim\limits_{t\to\infty}\left(6-4e^{-t}\right)=6.$

For particle Q, $\lim\limits_{t\to\infty}\left(3-\frac{1}{t}\right)=3.$

Because $6>3$, particle *P* will eventually be farther from the origin.

8-2b

75. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	0.3	1.7	2.8	4
$v_P(t)$ (meters per hour)	0	55	-29	55	48

The velocity of a particle, P, moving along the x-axis is given by the differentiable function v_P, where $v_P(t)$ is measured in meters per hour and t is measured in hours. Selected values of $v_P(t)$ are shown in the table above. Particle P is at the origin at time $t = 0$.

a) Justify why there must be at least one time t, for $0.3 \le t \le 2.8$, at which $v_P{}'(t)$, the acceleration of particle P, equals 0 meters per hour per hour.

b) Use a trapezoidal sum with the three subintervals $[0, 0.3]$, $[0.3, 1.7]$, and $[1.7, 2.8]$ to approximate the value of $\int_0^{2.8} v_P(t)\,dt$.

c) A second particle, Q, also moves along the x-axis so that its velocity for $0 \le t \le 4$ is given by $v_Q(t) = 45\sqrt{t}\cos\left(0.063t^2\right)$ meters per hour. Find the time interval during which the velocity of particle Q is at least 60 meters per hour. Find the distance traveled by particle Q during the interval when the velocity of particle Q is at least 60 meters per hour.

d) At time $t = 0$, particle Q is at position $x = -90$. Using the result from part (b) and the function v_Q from part (c), approximate the distance between particles P and Q at time $t = 2.8$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-2b

The first point is earned for expressing $v_P(2.8) - v_P(0.3) = 0$. Because these values come from the table, expressing $55 - 55$ earns the first point.

The second point is earned for declaring v_P to be continuous and connecting $v_P'(t)$ to the work for the first point.

Another approach involves an Extreme Value Theorem argument. The first point is earned for declaring $v_P(0.3) > v_P(1.7)$ and $v_P(1.7) < v_P(2.8)$. The second point is earned for declaring v_P to be continuous, and using continuity to conclude that $v_P(t)$ must have a minimum at some point c, $0.3 < c < 2.8$, so $v_P'(c) = 0$.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $v_P(2.8) - v_P(0.3) = 0$
- ☐ justification, using Mean Value Theorem

v_P is differentiable $\Rightarrow v_P$ is continuous on $0.3 \le t \le 2.8$.

$$\frac{v_P(2.8) - v_P(0.3)}{2.8 - 0.3} = \frac{55 - 55}{2.5} = 0$$

By the Mean Value Theorem, there is a value c, $0.3 < c < 2.8$, such that $v_P'(c) = 0$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Simplification is not required to earn this point. The point requires substitution of function values from the table.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ · answer, using trapezoidal sum

Solution:

8-2b

$$\int_0^{2.8} v_P(t)\,dt \approx 0.3\left(\frac{v_P(0)+v_P(0.3)}{2}\right) + 1.4\left(\frac{v_P(0.3)+v_P(1.7)}{2}\right)$$
$$+ 1.1\left(\frac{v_P(1.7)+v_P(2.8)}{2}\right)$$
$$= 0.3\left(\frac{0+55}{2}\right) + 1.4\left(\frac{55+(-29)}{2}\right) + 1.1\left(\frac{-29+55}{2}\right)$$
$$= 40.75$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for an open or closed interval. The interval must be stated explicitly analytically or verbally; using the endpoints of the interval as the limits of integration in a definite integral is not sufficient to earn the first point.

The second point is earned for a definite integral of the form $\int_{t_0}^{t_1} v_Q(t)\,dt$ or $\int_{t_0}^{t_1} |v_Q(t)|\,dt$.

The third point does not require units.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ interval
- ☐ definite integral
- ☐ distance

Solution:

$v_Q(t) = 60 \Rightarrow t = A = 1.866181$ or $t = B = 3.519174$

$v_Q(t) \geq 60$ for $A \leq t \leq B$

$$\int_A^B |v_Q(t)|\,dt = 106.108754$$

The distance traveled by particle Q during the interval $A \leq t \leq B$ is 106.109 (or 106.108) meters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-2b

The first point is earned for the integral $\int_0^{2.8} v_Q(t)\, dt$ or $\int_0^{2.8} |v_Q(t)|\, dt$.

The third point is earned for the difference between $x_P(2.8)$ and $x_Q(2.8)$. Any incorrect value from Part B of this question, handled correctly, makes the response eligible for the third point. The third point does not require units.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_0^{2.8} v_Q(t)\, dt$
- ☐ position of particle Q
- ☐ answer

Solution:

From Part B, the position of particle P at time $t = 2.8$ is $x_P(2.8) = \int_0^{2.8} v_P(t)\, dt \approx 40.75$.

$$x_Q(2.8) = x_Q(0) + \int_0^{2.8} v_Q(t)\, dt = -90 + 135.937653 = 45.937653$$

Therefore at time $t = 2.8$, particles P and Q are approximately $45.937653 - 40.75 = 5.188\ (\text{or } 5.187)$ meters apart.

8-2b

76. A particle moves along the x-axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

(a) Write an equation for the velocity $v(t)$ of the particle.

(b) Write an equation for the position $x(t)$ of the particle.

(c) For what values of t, $0 \leq t \leq \pi$, is the particle at rest?

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4\cos 2t\,dt \\ \therefore v(t) &= 2\sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2\sin 2t + 1 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4\cos 2t\,dt \\ \therefore v(t) &= 2\sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2\sin 2t + 1 \end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

8-2b

$x(t) = \int 2\sin 2t + 1 \text{ dt}$

$\therefore x(t) = -\cos 2t + t + C$

$x(0) = 0 \Rightarrow C = 1$

$\therefore x(t) = -\cos 2t + t + 1$

0	1	2	3

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$x(t) = \int 2\sin 2t + 1 \text{ dt}$

$\therefore x(t) = -\cos 2t + t + C$

$x(0) = 0 \Rightarrow C = 1$

$\therefore x(t) = -\cos 2t + t + 1$

Part C

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$
$$\sin 2t = -\tfrac{1}{2}$$
$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$
$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

0	1	2	3

The student response earns all of the following points:

8-2b

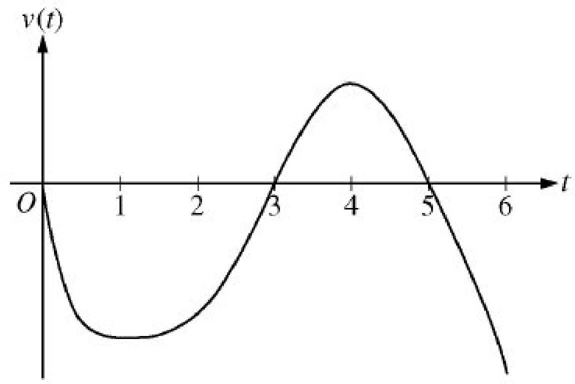
1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$
$$\sin 2t = -\frac{1}{2}$$
$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$
$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

8-2b

77.



Graph of v

A particle moves along the x-axis so that its velocity at time t, for $0 \le t \le 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t-axis and the graph of v on the intervals [0,3], [3,5], and [5,6] are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

(a) For $0 \leq t \leq 6$, find *both the time and the position of the particle when the particle is farthest to the left. Justify your answer.*

(b) For how many values of t, where $0 \le t \le 6$, is the particle at $x = -8$? Explain your reasoning.

(c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.

(d) *During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.*

Part A

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t)dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t)dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t)dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

8-2b

0	1	2	3

The student response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t)dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t)dt = -2 - 8 = -10$$
$$x(6) = -2 + \int_0^6 v(t)dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

0	1	2	3

The student response earns all of the following points:

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

8-2b

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Part C

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

0	1

The student response earns all of the following points:

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

0	1	2

The student response earns all of the following points:

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

8-2b

78. A particle moves along the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.

(a) Write a polynomial expression for the position of the particle at any time $t \geq 0$.

(b) For what values of t, $0 \leq t \leq 3$, is the particle's instantaneous velocity the same as its average velocity on the closed interval [0,3]?

(c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$x(t) = \int v(t)dt = \int (3t^2 - 2t - 1)dt$$
$$= t^3 - t^2 - t + C$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5;\ C = 3$$
$$x(t) = t^3 - t^2 - t + 3$$

0	1	2	3 ✓

The student response earns all of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

8-2b

$$x(t) = \int v(t)dt = \int (3t^2 - 2t - 1)dt$$
$$= t^3 - t^2 - t + C$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5;\ C = 3$$
$$x(t) = t^3 - t^2 - t + 3$$

Part B

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3)-x(0)}{3-0}$$
$$= \frac{18-3}{3} = 5$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 =$ an avg. velocity in [0, 3]

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the correct average velocity = 5

$$\text{avg. vel.} = \frac{x(3)-x(0)}{3-0}$$
$$= \frac{18-3}{3} = 5$$

8-2b

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 =$ an avg. velocity in [0, 3]

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

Part C

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\begin{aligned} \text{distance} &= \int_0^3 |v(t)|dt \\ &= \int_0^3 \left|3t^2 - 2t - 1\right|dt = 17 \end{aligned}$$

or

$$\begin{aligned} &v(t) = 3t^2 - 2t - 1 = 0 \\ &t = -\tfrac{1}{3}, t = 1 \\ &x(0) = 3 \\ &x(1) = 1 - 1 - 1 + 3 = 2 \\ &x(3) = 27 - 9 - 3 + 3 = 18 \end{aligned}$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

distance = $(3 - 2) + (18 - 2) = 17$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

8-2b

$$\text{distance} = \int_0^3 |v(t)|dt$$
$$= \int_0^3 \left|3t^2 - 2t - 1\right|dt = 17$$

or

$v(t) = 3t^2 - 2t - 1 = 0$
$t = -\frac{1}{3}, t = 1$
$x(0) = 3$
$x(1) = 1 - 1 - 1 + 3 = 2$
$x(3) = 27 - 9 - 3 + 3 = 18$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

distance = (3 − 2) + (18 − 2) = 17

8-2b

79. A particle moves along the x-axis so that its velocity at any time $t \geqq 0$ is given by $v(t) = 1 - \sin(2\pi t)$.

*(a) F*ind the acceleration $a(t)$ of the particle at any time t.

(b) Find all values of t, $0 \leqq t \leqq 2$, for which the particle is at rest.

(c) Find the position $x(t)$ of the particle at any time t if x(0) = 0.

Part A

2 points are earned for correct differentiation of velocity

$$a(t) = v'(t)$$
$$= -2\pi\cos(2\pi t)$$

0	1	2

The student response earns all of the following points:

2 points are earned for correct differentiation of velocity

$$a(t) = v'(t)$$
$$= -2\pi\cos(2\pi t)$$

Part B

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$v(t) = 0$

$1 - \sin(2\pi t) = 0$ or $1 = \sin(2\pi t)$

$2\pi t = \frac{\pi}{2} + 2k\pi,$

8-2b

where $k = 0, \pm 1, \pm 2, \ldots$

and $0 \leqq t \leqq 2$

$\therefore t = \frac{1}{4}$ and $t = \frac{5}{4}$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$v(t) = 0$

$1 - \sin(2\pi t) = 0$ or $1 = \sin(2\pi t)$

$2\pi t = \frac{\pi}{2} + 2k\pi,$

where $k = 0, \pm 1, \pm 2, \ldots$

and $0 \leqq t \leqq 2$

$\therefore t = \frac{1}{4}$ and $t = \frac{5}{4}$

Part C

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

8-2b

1 point is earned for finding value of C

$$
\begin{aligned}
x(t) &= \int v(t)\mathrm{dt} \\
&= \int [\,1 - \sin(2\pi t)]\ \mathrm{dt} \\
&= t + \tfrac{1}{2\pi}\cos(2\pi t) + C
\end{aligned}
$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi}\cos(2\pi t) - \frac{1}{2\pi}$$

0	1	2	3	4 ✓

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$
\begin{aligned}
x(t) &= \int v(t)\mathrm{dt} \\
&= \int [\,1 - \sin(2\pi t)]\ \mathrm{dt} \\
&= t + \tfrac{1}{2\pi}\cos(2\pi t) + C
\end{aligned}
$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi}\cos(2\pi t) - \frac{1}{2\pi}$$

8-2b

80. A particle moves along the x-axis so that its velocity at time t is given by

$$v(t) = -(t+1)\sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.

(a) Find the acceleration of the particle at time $t = 2$. Is the speed of the particle increasing at $t = 2$? Why or why not?

(b) Find all times t in the open interval $0 < t < 3$ when the particle changes direction. Justify your answer.

(c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

(d) During the time interval $0 \le t \le 3$, what is the greatest distance between the particle and the origin? Show the work that leads to your answer.

Part A

1 point is earned for correctly solving for $a(2)$

$a(2) = v'(2) = 1.587 \text{ or } 1.588$
$v(2) = -3\sin(2) < 0$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for correctly solving for $a(2)$

$a(2) = v'(2) = 1.587 \text{ or } 1.588$
$v(2) = -3\sin(2) < 0$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.

Part B

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

8-2b

$v(t) = 0$ when $\dfrac{t^2}{2} = \pi$

$t = \sqrt{2\pi}$ or 2.506 or 2.507

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.

0	1	2

The student response earns all of the following points:

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

$v(t) = 0$ when $\dfrac{t^2}{2} = \pi$

$t = \sqrt{2\pi}$ or 2.506 or 2.507

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

Distance $= \displaystyle\int_0^3 |v(t)|dt = 4.333$ or 4.334

0	1	2	3

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

8-2b

$$\text{Distance} = \int_0^3 |v(t)|dt = 4.333 \text{ or } 4.334$$

Part D

1 point is earned for the ± distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t)dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t)dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the ± distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t)dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t)dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

8-2b

81. A particle, initially at rest, moves along the x-axis so that its acceleration at any time $t \geq 0$ is given by $a(t) = 12t^2 - 4$. The position of the particle when $t = 1$ is $x(1) = 3$.

(a) Find the values of t for which t*he particle is at rest.*

(b) Write an expression for the position $x(t)$ of the particle at any time $t \geq 0$.

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

2 points are earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$$
\begin{aligned}
v(t) &= 4t^3 - 4t \\
v(t) &= 4t^3 - 4t = 0 \\
&\quad 4t\left(t^2 - 1\right) = 0 \\
\therefore t &= 0,\ t = 1
\end{aligned}
$$

0	1	2	3	4 ✓

The student response earns all of the following points:

2 points are earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$$
\begin{aligned}
v(t) &= 4t^3 - 4t \\
v(t) &= 4t^3 - 4t = 0 \\
&\quad 4t\left(t^2 - 1\right) = 0 \\
\therefore t &= 0,\ t = 1
\end{aligned}
$$

Part B

8-2b

2 point are earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(t) = 1^4 - 2 \cdot 1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

0	1	2	3

The student response earns all of the following points:

2 point are earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(t) = 1^4 - 2 \cdot 1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

Part C

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

8-2b

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	
Total Distance		$\boxed{10}$

0	1	2

The student response earns all of the following points:

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	
Total Distance		$\boxed{10}$

8-2b

82. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

(a) What is the acceleration of the object at time $t = 4$?

(b) Consider the following two statements.

Statement *I*: For $3 < t < 4.5$, the velocity of the object is decreasing.

Statement II: For $3 < t < 4.5$, the speed of the object is increasing.

Are either or both of these statements correct? For each statement provide a reason why it is correct or not correct.

(c) What is the total distance traveled by the object over the time interval $0 \leq t \leq 4$?

(d) What is the position of the object at time $t = 4$?

Part A

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

✓

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

Part B

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

8-2b

$$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

Part C

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handle change of direction at student's turning point answer

1 point is earned for the answer

$$\text{Distance} = \int_0^4 |v(t)|dt = 2.387$$

OR

8-2b

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(0) = 2$
$x(4) = 2 + \dfrac{9}{2\pi} = 3.43239$
$v(t) = 0 \text{ When } t = 3$
$x(3) = \frac{6}{\pi} + 2 = 3.90986$
$|x\,(3) - x(0)| + |x\,(4) - x(3)| = \dfrac{15}{2\pi} = 2.387$

0	1	2	3

The student response earns all of the following points:

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v\,(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handles change of direction at student's turning point answer

1 point is earned for the answer

$$\text{Distance} = \int_0^4 |v\,(t)|dt = 2.387$$

OR

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(0) = 2$
$x(4) = 2 + \dfrac{9}{2\pi} = 3.43239$
$v(t) = 0 \text{ When } t = 3$
$x(3) = \frac{6}{\pi} + 2 = 3.90986$
$|x\,(3) - x(0)| + |x\,(4) - x(3)| = \dfrac{15}{2\pi} = 2.387$

Part D

8-2b

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$$

OR

$$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$
$$x(4) = 2 + \frac{9}{2\pi} = 3.432$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$$

OR

8-2b

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$

$x(4) = 2 + \frac{9}{2\pi} = 3.432$

8-2b

83. A particle moves along the x-axis with velocity given by $v(t) = \frac{10\sin(0.4t^2)}{t^2-t+3}$ for time $0 \le t \le 3.5$. The particle is at position $x = -5$ at time $t = 0$.

(a) Find the acceleration of the particle at time $t = 3$.

(b) Find the position of the particle at time $t = 3$.

(c) Evaluate $\int_0^{3.5} v(t)\,dt$, and evaluate $\int_0^{3.5} |v(t)|\,dt$. Interpret the meaning of each integral in the context of the problem.

(d) A second particle moves along the x-axis with position given by $x_2(t) = t^2 - t$ for $0 \le t \le 3.5$. At what time t are the two particles moving with the same velocity?

Part A

1 point is earned for answer

$v'(3) = -2.118$

The acceleration of the particle at time $t = 3$ is -2.118.

0	1

The student earns all of the following points:

1 point is earned for answer

$v'(3) = -2.118$

The acceleration of the particle at time $t = 3$ is -2.118.

Part B

1 point is earned for $\int_0^3 v(t)\,dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t)\,dt = -5 + \int_0^3 v(t)\,dt = -1.760213$$

8-2b

The position of the particle at time $t = 3$ is -1.760.

0	1	2	3

The student earns all of the following point:

1 point is earned for $\int_0^3 v(t)\,dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t)\,dt = -5 + \int_0^3 v(t)\,dt = -1.760213$$

The position of the particle at time $t = 3$ is -1.760.

Part C

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t)\,dt$ and $\int_0^{3.5} |v(t)|\,dt$

$$\int_0^{3.5} v(t)\,dt = 2.844 \text{ (or } 2.843\text{)}$$

$$\int_0^{3.5} |v(t)|\,dt = 3.737$$

The integral $\int_0^{3.5} v(t)\,dt$ is the displacement of the particle over the time interval $0 \le t \le 3.5$.

8-2b

The integral $\int_0^{3.5} |v(t)|\,dt$ is the total distance traveled by the particle over the time interval $0 \le t \le 3.5$.

0	1	2	3

The student earns all of the following point:

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t)\,dt$ and $\int_0^{3.5} |v(t)|\,dt$

$$\int_0^{3.5} v(t)\,dt = 2.844 \text{ (or } 2.843)$$

$$\int_0^{3.5} |v(t)|\,dt = 3.737$$

The integral $\int_0^{3.5} v(t)\,dt$ is the displacement of the particle over the time interval $0 \le t \le 3.5$.

The integral $\int_0^{3.5} |v(t)|\,dt$ is the total distance traveled by the particle over the time interval $0 \le t \le 3.5$.

Part D

1 point is earned if sets $v(t) = x_2{}'(t)$

1 point is earned for the answer

$$v(t) = x_2{}'(t)$$

$v(t) = 2t - 1 \Rightarrow t = 1.57054$

8-2b

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

0	1	2

The student earns all of the following point:

1 point is earned if sets $v(t) = {x_2}'(t)$

1 point is earned for the answer

$v(t) = {x_2}'(t)$

$v(t) = 2t - 1 \Rightarrow t = 1.57054$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

8-2b

84. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle, P, is moving along the x-axis. The velocity of particle P at time t is given by $v_P(t) = \sin\left(t^{1.5}\right)$ for $0 \le t \le \pi$. At time $t = 0$, particle P is at position $x = 5$.

A second particle, Q, also moves along the x-axis. The velocity of particle Q at time t is given by $v_Q(t) = (t - 1.8) \cdot 1.25^t$ for $0 \le t \le \pi$. At time $t = 0$, particle Q is at position $x = 10$.

(a) Find the positions of particles P and Q at time $t = 1$.

(b) Are particles P and Q moving toward each other or away from each other at time $t = 1$? Explain your reasoning.

(c) Find the acceleration of particle Q at time $t = 1$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Explain your reasoning.

(d) Find the total distance traveled by particle P over the time interval $0 \le t \le \pi$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the explicit presentation of at least one definite integral, either $\int_0^1 v_P(t)\,dt$ or $\int_0^1 v_Q(t)\,dt$.

The first point must be earned to be eligible for the second and third points.

The second point is earned for adding the initial condition to at least one of the definite integrals and finding the correct position.

8-2b

Writing $\int_0^1 v_P(t) + 5 = 5.370660$ does not earn a position point, because the missing dt makes this statement unclear or false. However, $5 + \int_0^1 v_P(t) = 5.370660$ does earn the position point because it is not ambiguous. Similarly, for the position of Q.

Read unlabeled answers presented left to right, or top to bottom, as $x_P(1)$ and $x_Q(1)$, respectively.

Special case 1: A response of $x_P(1) = 5 + \int_0^a v_P(t)dt = 5.370660$ AND $x_Q(1) = 10 + \int_0^a v_Q(t)dt = 8.564355$ for $a \neq 1$ earns one point.

Special case 2: A response of $x_P(1) = 5 + \int v_P(t)dt = 5.370660$ AND $x_Q(1) = 10 + \int v_Q(t)dt = 8.564355$ or the equivalent, never providing the definite integrals, earns one point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $x_P(1)$ is 5.007 (or 5.006).

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ One definite integral
- ☐ One position
- ☐ The other position

Solution:

$$x_P(1) = 5 + \int_0^1 v_P(t)dt = 5.370660$$

At time $t = 1$, the position of particle P is $x = 5.371$ (or 5.370).

$$x_Q(1) = 10 + \int_0^1 v_Q(t)dt = 8.564355$$

At time $t = 1$, the position of particle Q is $x = 8.564$.

Part B

8-2b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the sign of $v_P(1)$ or $v_Q(1)$ to determine the direction of motion for one of the particles. This point cannot be earned without reference to the sign of $v_P(1)$ or $v_Q(1)$.

It is not necessary to present an explicit value for $v_P(1)$ or $v_Q(1)$, but if a value is presented, it must be correct as far as reported, up to three places after the decimal.

Read with imported incorrect position values from part (a).

If one or both position values were not found in part (a), but are found in part (b), the points for part (a) are not earned retroactively.

To earn the second point the explanation must be based on the signs of $v_P(1)$ and $v_Q(1)$ and the relative positions of particle P and particle Q at $t = 1$. References to other values of time, such as $t = 0$, are not sufficient.

Degree mode: $v_P(1) = 0.017$. (See degree mode statement in part (a).)

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Direction of motion for one particle
- ☐ Answer with explanation

Solution:

$v_P(1) = \sin\left(1^{1.5}\right) = 0.841471 > 0$

At time $t = 1$, particle P is moving to the right.

$v_Q(1) = (1 - 1.8) \cdot 1.25^1 = -1 < 0$

At time $t = 1$, particle Q is moving to the left.

At time $t = 1$, $x_P(1) < x_Q(1)$, so particle P is to the left of particle Q.

Thus, at time $t = 1$, particles P and Q are moving toward each other.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the acceleration must be explicitly connected to v'_Q (e.g., $v'_Q(1) = 1.026856$).

8-2b

The first point is not earned for an unsupported value of 1.027 (or 1.026). The setup, $v'_Q(1)$, must be shown. Presenting only $a_Q(1) = 1.027$ (or 1.026) without indication that $v'_Q = a_Q$ is not enough to earn the first point.

A response does not need to present a value for $v_Q\ (1)$; the sign is sufficient.

To earn the second point a response must compare the signs of a_Q and v_Q at $t = 1$. Considering only one sign is not sufficient.

After the first point has been earned, a response declaring only "velocity and acceleration are of opposite signs at $t = 1$ so the particle is slowing down" (or equivalent) earns the second point.

The second point may be earned without the first, as long as the response does not present an incorrect value or sign for $v_Q(1)$ and concludes the particle is slowing down because velocity and acceleration have opposite signs at $t = 1$.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Setup and acceleration
- [ ] Speed decreasing with reason

Solution:

$a_Q(1) = v'_Q(1) = 1.026856$

The acceleration of particle Q is 1.027 (or 1.026) at time $t = 1$.

$v_Q(1) = -1 < 0$ and $a_Q(1) > 0$

The speed of particle Q is decreasing at time $t = 1$ because the velocity and acceleration have opposite signs.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $\displaystyle\int_0^{\pi} |v_P(t)|dt$.

The first point can also be earned for a sum (or difference) of definite integrals, such as a $\displaystyle\int_0^{2.145029} v_P(t)dt - \int_{2.145029}^{\pi} v_P(t)dt$, provided the response has indicated $v_P\,(2.145029) = 0$.

The second point can only be earned for the correct answer.

8-2b

The unsupported value 1.931 earns no points.

A response reporting the distance traveled by particle Q as $\int_0^{\pi} |v_Q(t)|dt = 3.506$ earns the first point and is not eligible for the second point.

In degree mode, the total distance traveled is 0.122. (See degree mode statement in part (a).) In the degree mode case, the response must present $\int_0^{\pi} |v_P(t)|dt$ in order to earn the first point because $\int_0^{\pi} |v_P(t)|dt = \int_0^{\pi} v_P(t)dt.$

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\int_0^{\pi} |v_P(t)|dt = 1.93148$$

Over the time interval $0 \leq t \leq \pi$, the total distance traveled by particle P is 1.931.

8-2b

85. A particle moves on the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 12t^2 - 36t + 15$. At $t = 1$, the particle is at the origin.

(a) Find the position $x(t)$ of the particle at any time $t \geq 0$.

(b) Find all values of t for which the particle is at rest.

(c) Find the maximum velocity of the particle for $0 \leq t \leq 2$.

Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$$x(t) = 4t^3 - 18t^2 + 15t + C$$

$0 = x(1) = 4 - 18 + 15 + C$

$\therefore C = -1$

$$\boxed{x(t) = 4t^3 - 18t^2 + 15t - 1}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$$x(t) = 4t^3 - 18t^2 + 15t + C$$

$0 = x(1) = 4 - 18 + 15 + C$

$\therefore C = -1$

$$\boxed{x(t) = 4t^3 - 18t^2 + 15t - 1}$$

Part B

8-2b

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$0 = v(t) = 12t^2 - 36t + 15$

$3(2t - 1)(2t - 5) = 0$

$\boxed{t = \frac{1}{2}, \frac{5}{2}}$

0	1	2

The student response earns all of the following points:

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$0 = v(t) = 12t^2 - 36t + 15$

$3(2t - 1)(2t - 5) = 0$

$\boxed{t = \frac{1}{2}, \frac{5}{2}}$

Part C

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

8-2b

$$\frac{dv}{dt} = 24t - 36$$
$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$
$$v(0) = 15$$
$$v\left(\frac{3}{2}\right) = -12$$
$$v(2) = -9$$
$$\boxed{\text{Maximum velocity is 15}}$$

0	1	2

The student response earns all of the following points:

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

$$\frac{dv}{dt} = 24t - 36$$
$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$
$$v(0) = 15$$
$$v\left(\frac{3}{2}\right) = -12$$
$$v(2) = -9$$
$$\boxed{\text{Maximum velocity is 15}}$$

Part D

1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\text{Total Distance} = \int_0^{1/2} v(t)dt - \int_{1/2}^{2} v(t)dt$$
$$= \left(x\left(\frac{1}{2}\right) - x(0)\right) - \left(x(2) - x\left(\frac{1}{2}\right)\right)$$
$$= \frac{5}{2} - (-1) - \left(-11 - \frac{5}{2}\right) = \boxed{17}$$

8-2b

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\begin{aligned}\text{Total Distance} &= \int_0^{1/2} v(t)dt - \int_{1/2}^{2} v(t)dt \\ &= \left(x\left(\tfrac{1}{2}\right) - x(0)\right) - \left(x\left(2\right) - x\left(\tfrac{1}{2}\right)\right) \\ &= \tfrac{5}{2} - (-1) - \left(-11 - \tfrac{5}{2}\right) = \boxed{17}\end{aligned}$$

8-2b

86. A particle moves along the x-axis so that its velocity v at time t, for $0 \leq t \leq 5$, is given by $v(t) = \text{In}(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

(a) Find the acceleration of the particle at time $t = 4$.

(b) Find all times t in the open interval $0 < t < 5$ at which the particle changes direction. During which time intervals, for $0 \leq t \leq 5$, does the particle travel to the left?

(c) Find the position of the particle at time $t = 2$.

(d) Find the average speed of the particle over the interval $0 \leq t \leq 2$.

Part A

1 point is earned for the answer

$a(4) = v'(4) = \frac{5}{7}$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$a(4) = v'(4) = \frac{5}{7}$

Part B

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$v(t) = 0$

$$t^2 - 3t + 3 = 1$$
$$t^2 - 3t + 2 = 0$$

$(t - 2)(t - 1) = 0$

$t = 1, 2$

$v(t) > 0$ for $0 < t < 1$

$v(t) < 0$ for $1 < t < 2$

8-2b

$v(t) > 0$ for $2 < t < 5$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$v(t) = 0$

$$t^2 - 3t + 3 = 1$$
$$t^2 - 3t + 2 = 0$$

$(t - 2)(t - 1) = 0$

$t = 1, 2$

$v(t) > 0$ for $0 < t < 1$

$v(t) < 0$ for $1 < t < 2$

$v(t) > 0$ for $2 < t < 5$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

Part C

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

8-2b

$$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$$
$$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$$

= 8.368 or 8.369

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$$
$$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$$

= 8.368 or 8.369

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2}\int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$

0	1	2

8-2b

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2}\int_{0}^{2}|v\left(t\right)|dt=0.370\text{ or }0.371$$

8-2b

87. A particle moves along a straight line. For $0 \le t \le 5$, the velocity of the particle is given by $v(t) = -2 + (t^2 + 3t)^{6/5} - t^3$, and the position of the particle is given by $s(t)$. It is known that $s(0) = 10$.

(a) Find all values of t in the interval $2 \le t \le 4$ for which the speed of the particle is 2.

(b) Write an expression involving an integral that gives the position $s(t)$. Use this expression to find the position of the particle at time $t = 5$.

(c) Find all times t in the interval $0 \le t \le 5$ at which the particle changes direction. Justify your answer.

(d) Is the speed of the particle increasing or decreasing at time $t = 4$? Give a reason for your answer.

Part A

1 point is earned for considering $|v(t)| = 2$

1 point is earned for the answer

Solve $|v(t)| = 2$ on $2 \le t \le 4$.

$t = 3.128$ (or 3.127) and $t = 3.473$

0	1	2

The student response earns all of the following points:

1 point is earned for considering $|v(t)| = 2$

1 point is earned for the answer

Solve $|v(t)| = 2$ on $2 \le t \le 4$.

$t = 3.128$ (or 3.127) and $t = 3.473$

Part B

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x)dx$$

$$s(5) = 10 + \int_0^5 v(x)dx = -9.207$$

8-2b

0	1	2

The student response earns all of the following points:

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x)dx$$
$$s(5) = 10 + \int_0^5 v(x)dx = -9.207$$

Part C

1 point is earned for considering $v(t) = 0$

2 points are earned for answering with justification

$v(t) = 0$ when $t = 0.536033, 3.317756$

$v(t)$ changes sign from negative to positive at time $t = 0.536033$.

$v(t)$ changes sign from positive to negative at time $t = 3.317756$.

Therefore, the particle changes direction at time $t = 0.536$ and time $t = 3.318$ (or 3.317).

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

2 points are earned for answering with justification

$v(t) = 0$ when $t = 0.536033, 3.317756$

$v(t)$ changes sign from negative to positive at time $t = 0.536033$.

$v(t)$ changes sign from positive to negative at time $t = 3.317756$.

8-2b

Therefore, the particle changes direction at time $t = 0.536$ and time $t = 3.318$ (or 3.317).

Part D

2 points are earned for the conclusion with reason

$v(4) = -11.475758 < 0$, $a(4) = v'(4) = -22.295714 < 0$

The speed is increasing at time $t = 4$ because velocity and acceleration have the same sign.

0	1	2 ✓

The student response earns all of the following points:

2 points are earned for the conclusion with reason

$v(4) = -11.475758 < 0$, $a(4) = v'(4) = -22.295714 < 0$

The speed is increasing at time $t = 4$ because velocity and acceleration have the same sign.

8-2b

88. For $0 \leq t \leq 12$, a particle moves along the x-axis. The velocity of the particle at time t is given by $v(t) = \cos\left(\frac{\pi}{6}t\right)$. The particle is at position $x = -2$ at time $t = 0$.

(a) For $0 \leq t \leq 12$, when is the particle moving to the left?

(b) Write, but do not evaluate, an integral expression that gives the total distance traveled by the particle from time $t = 0$ to time $t = 6$.

(c) Find the acceleration of the particle at time t. Is the speed of the particle increasing, decreasing, or neither at time $t = 4$? Explain your reasoning.

(d) Find the position of the particle at time $t = 4$.

Part A

1 point is earned for considering $v(t) = 0$

1 point is earned for the interval

$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$.

0	1	2

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

1 point is earned for the interval

$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$.

Part B

1 point is earned for the answer

$$\int_0^6 |v(t)|dt$$

8-2b

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$\int_0^6 |v(t)|dt$$

Part C

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6}\sin\left(\frac{\pi}{6}t\right)$$
$$a(4) = -\frac{\pi}{6}\sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$
$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6}\sin\left(\frac{\pi}{6}t\right)$$
$$a(4) = -\frac{\pi}{6}\sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$
$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.

8-2b

Part D

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$x(4) = -2 + \int_0^4 \cos\left(\frac{\pi}{6}t\right) dt$$
$$= -2 + \left[\tfrac{6}{\pi}\sin\left(\tfrac{\pi}{6}t\right)\right]_0^4$$
$$= -2 + \tfrac{6}{\pi}\left[\sin\left(\tfrac{2\pi}{3}\right) - 0\right]$$
$$= -2 + \tfrac{6}{\pi}\cdot\frac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$x(4) = -2 + \int_0^4 \cos\left(\frac{\pi}{6}t\right) dt$$
$$= -2 + \left[\tfrac{6}{\pi}\sin\left(\tfrac{\pi}{6}t\right)\right]_0^4$$
$$= -2 + \tfrac{6}{\pi}\left[\sin\left(\tfrac{2\pi}{3}\right) - 0\right]$$
$$= -2 + \tfrac{6}{\pi}\cdot\frac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}$$

8-2b

89. For $0 \le t \le 6$, a particle is moving along the x-axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t) = 2\sin(e^{t/4}) + 1$. The acceleration of the particle is given by $a(t) = 1/2e^{t/4}\cos(e^{t/4})$ and $x(0) = 2$.

(a) Is the speed of the particle increasing or decreasing at time $t = 5.5$? Give a reason for your answer.

(b) Find the average velocity of the particle for the time period $0 \le t \le 6$.

(c) *Find the total distance traveled by the particle from tim*e $t = 0$ to $t = 6$.

(d) For $0 \le t \le 6$, the particle changes direction exactly once. Find the position of the particle at that time.

Part A

2 points are earned for conclusion with reason

$v(5.5) = -0.45337$, $a(5.5) = -1.35851$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.

0	1	2

The student response earns all of the following points:

2 points are earned for conclusion with reason

$v(5.5) = -0.45337$, $a(5.5) = -1.35851$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.

Part B

1 point is earned for integral

1 point is earned for the answer

Average velocity $= \frac{1}{6}\int_0^6 v(t)dt = 1.949$

0	1	2

The student response earns all of the following points:

1 point is earned for integral

8-2b

1 point is earned for the answer

Average velocity $= \frac{1}{6}\int_0^6 v(t)dt = 1.949$

Part C

1 point is earned for integral

1 point is earned for the answer

Distance $= \int_0^6 |v(t)|dt = 12.573$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

Distance $= \int_0^6 |v(t)|dt = 12.573$

Part D

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$x(b) = 2 + \int_0^b v(t)dt = 14.134$ or 14.135

0	1	2	3 ✓

8-2b

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

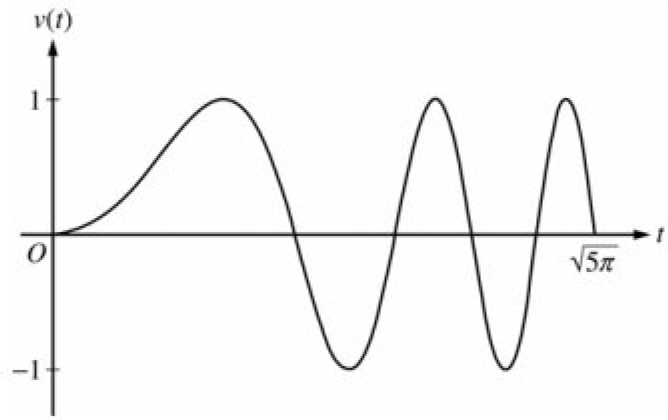
$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t)dt = 14.134 \text{ or } 14.135$$

8-2b

90.



A particle moves along the x-axis so that its velocity v at time $t \geq 0$ is given by $v(t) = \sin(t^2)$. The graph of v is shown above for $0 \leq t \leq \sqrt{5\pi}$. The position of the particle at time t is $x(t)$ and its position at time $t = 0$ is $x(0) = 5$.

(a) Find the acceleration of the particle at time $t = 3$.

(b) Find the total distance traveled by the particle from time $t = 0$ to $t = 3$.

(c) Find the position of the particle at time $t = 3$.

(d) For $0 \leq t \leq \sqrt{5\pi}$, find the time t at which the particle is farthest to the right. Explain your answer.

Part A

1 point is earned for $a(3)$

$a(3) = v'(3) = 6\cos 9 = -5.466$ or -5.467

0	1

The student response earns all of the following points:

1 point is earned for $a(3)$

$a(3) = v'(3) = 6\cos 9 = -5.466$ or -5.467

Part B

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)|dt = 1.702$$

8-2b

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

$x(0) = 5$

$$x\left(\sqrt{\pi}\right) = 5 + \int_0^{\sqrt{\pi}} v(t)dt = 5.89483$$
$$x\left(\sqrt{2\pi}\right) = 5 + \int_0^{\sqrt{2\pi}} v(t)dt = 5.43041$$
$$x(3) = 5 + \int_0^{3} v(t)dt = 5.77356$$
$$\left|x\left(\sqrt{\pi}\right) - x(0)\right| + \left|x\left(\sqrt{2\pi}\right) - x\left(\sqrt{\pi}\right)\right| + \left|x(3) - x\left(\sqrt{2\pi}\right)\right| = 1.702$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)|dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

$x(0) = 5$

8-2b

$$x\left(\sqrt{\pi}\right) = 5 + \int_0^{\sqrt{\pi}} v\left(t\right)dt = 5.89483$$

$$x\left(\sqrt{2\pi}\right) = 5 + \int_0^{\sqrt{2\pi}} v\left(t\right)dt = 5.43041$$

$$x\left(3\right) = 5 + \int_0^{3} v\left(t\right)dt = 5.77356$$

$$\left|x\left(\sqrt{\pi}\right) - x\left(0\right)\right| + \left|x\left(\sqrt{2\pi}\right) - x\left(\sqrt{\pi}\right)\right| + \left|x\left(3\right) - x\left(\sqrt{2\pi}\right)\right| = 1.702$$

Part C

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t)dt = 5.773 \text{ or } 5.774$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t)dt = 5.773 \text{ or } 5.774$$

Part D

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

8-2b

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t)dt$, the particle's positions at the times it changes from rightward to leftward movement are:

$T:$	0	$\sqrt{\pi}$	$\sqrt{3\pi}$	$\sqrt{5\pi}$
$x(T):$	5	5.895	5.788	5.752

The particle is farthest to the right when $T = \sqrt{\pi}$.

0	1	2	3

The student response earns all of the following points:

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t)dt$, the particle's positions at the times it changes from rightward to leftward movement are:

8-2b

$T:$	0	$\sqrt{\pi}$	$\sqrt{3\pi}$	$\sqrt{5\pi}$
$x(T):$	5	5.895	5.788	5.752

The particle is farthest to the right when $T = \sqrt{\pi}$.

8-2b

91. A particle moves on the x-axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$.

(a) Find the acceleration of the particle at $t = 0$.

(b) Find the velocity of the particle when its acceleration is 0.

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 5$.

Part A

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$
$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$
$$a(0) = x''(0) = -2 - 2 = -4$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$
$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$
$$a(0) = x''(0) = -2 - 2 = -4$$

Part B

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

8-2b

$$x''(t) = -2e^{-t}(2-t) = 0$$
$$t = 2$$
$$x'(2) = v(2) = 2e^{-2}(1-2)$$
$$= \frac{-2}{e^2}$$
$$\approx -0.271$$

0	1	2

The student response earns all of the following points:

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$x''(t) = -2e^{-t}(2-t) = 0$$
$$t = 2$$
$$x'(2) = v(2) = 2e^{-2}(1-2)$$
$$= \frac{-2}{e^2}$$
$$\approx -0.271$$

Part C

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and $x(\text{turning point})$

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

$$x'(t) = v(t) = 2e^{-t}(1-t) = 0$$
$$t = 1$$
$$x(0) = 0$$
$$x(1) = \frac{2}{e} \approx 0.736$$
$$x(5) = \frac{10}{e^5} \approx 0.067$$

8-2b

$$\text{Distance} = \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$$
$$\approx 1.404$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and x(turning point)

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

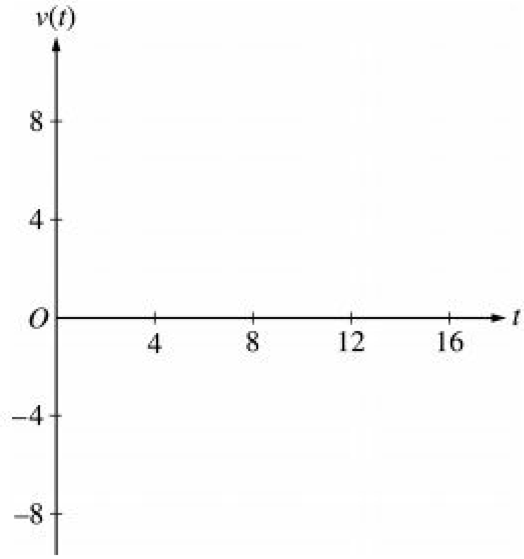
$$x'(t) = v(t) = 2e^{-t}(1-t) = 0$$
$$t = 1$$
$$x(0) = 0$$
$$x(1) = \frac{2}{e} \approx 0.736$$
$$x(5) = \frac{10}{e^5} \approx 0.067$$

$$\text{Distance} = \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$$
$$\approx 1.404$$

8-2b

92. A particle moves along the x-axis so that its velocity v at any time t, for $0 \leq t \leq 16$, is given by $v(t) = e^{2\sin t} - 1$. At time $t = 0$, the particle is at the origin.

(a) On the axes provided, sketch the graph of $v(t)$ for $0 \leq t \leq 16$.



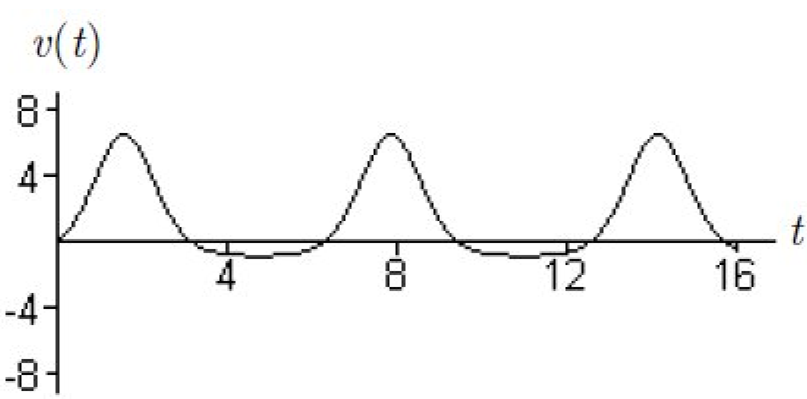
(b) During what intervals of time is the particle moving to the left? Give a reason for your answer.

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 4$.

(d) Is there any time t, $0 < t \leq 16$, at which the particle returns to the origin? Justify your answer.

Part A

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values

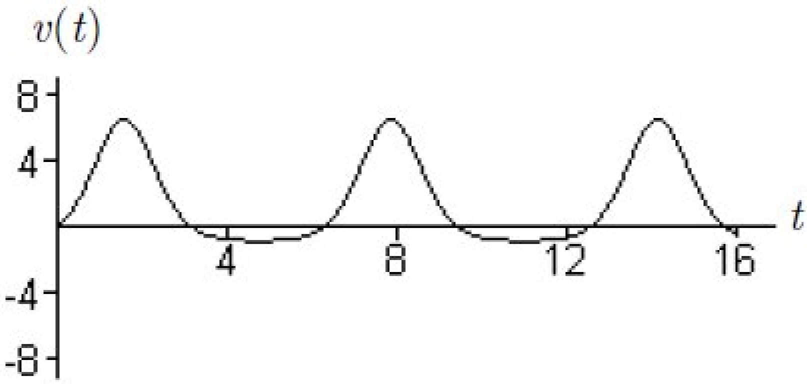


0	1

The student response earns all of the following points:

8-2b

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values



Part B

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$v(t) < 0,$ i.e. $e^{2\sin t} < 1.$
$(\pi, 2\pi),\ \ (3\pi, 4\pi)$and $(5\pi, 16]$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$v(t) < 0,$ i.e. $e^{2\sin t} < 1.$
$(\pi, 2\pi),\ \ (3\pi, 4\pi)$and $(5\pi, 16]$

8-2b

Part C

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

$$\int_0^4 |v(t)|dt = 10.542$$

or

$$v(t) = e^{2\sin t} - 1 = 0$$
$$t = 0 \text{ or } t = \pi$$
$$x(\pi) = \int_0^{\pi} v(t)dt = 10.10656$$
$$x(4) = \int_0^4 v(t)dt = 9.67066$$
$$|x(\pi) - x(0)| + |x(4) - x(\pi)|$$
$$= 10.542$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

8-2b

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$$x(4) = \int_0^4 v(t)dt = 9.67066$$
$$|x(\pi) - x(0)| + |x(4) - x(\pi)|$$
$$= 10.542$$

Part D

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t)dt > 0$ for all $T > 0$.

0	1	2

The student response earns all of the following points:

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t)dt > 0$ for all $T > 0$.

8-2b

93.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time t = 0 seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

(a) Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{10}^{70} v(t) dt$ in terms of the rockets flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t) dt$.

(c) Rocket B is launched upward with an acceleration of $a(t) = \frac{3}{\sqrt{t+1}}$ feet per second per second. At time t = 0 seconds, the initial height of the rocket is 0 feet, and the initial velocity is 2 feet per second. Which of the two rockets is traveling faster at time t = 80 seconds? Explain your answer.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0}=\frac{49-5}{80}=\frac{11}{20}\ \text{ft/sec}^2$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0}=\frac{49-5}{80}=\frac{11}{20}\ \text{ft/sec}^2$$

Part B

8-2b

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v\,(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$20[v(20) + v(40) + v(60)]$

$= 20[22 + 35 + 44] = 2020$ ft

0	1	2	3

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v\,(t)dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$20[v(20) + v(40) + v(60)]$

$= 20[22 + 35 + 44] = 2020$ ft

Part C

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

8-2b

Let $v_B(t)$ be the velocity of rocket B at time t.

$$v_B\left(t\right)=\int\frac{3}{\sqrt{t+1}}dt=6\sqrt{t+1}+C$$
$$2=v_B\left(0\right)=6+C$$
$$v_B\left(t\right)=6\sqrt{t+1}-4$$
$$v_B\left(80\right)=50>49=v_B\left(80\right)$$

Rocket B is traveling faster at time $t = 80$ seconds.

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

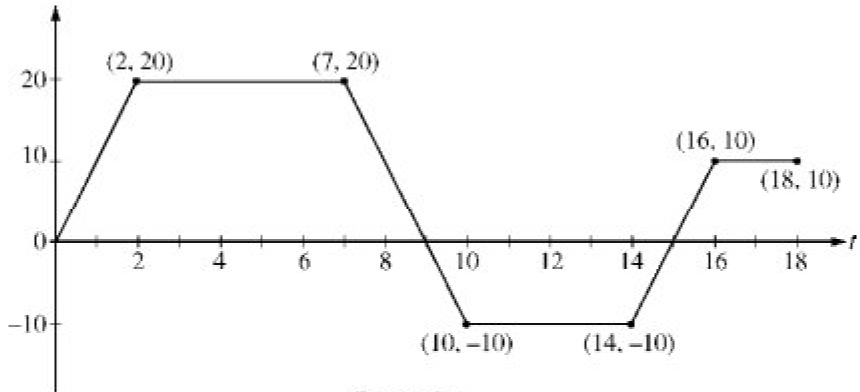
Let be the velocity of rocket B at time t.

$$v_B\left(t\right)=\int\frac{3}{\sqrt{t+1}}dt=6\sqrt{t+1}+C$$
$$2=v_B\left(0\right)=6+C$$
$$v_B\left(t\right)=6\sqrt{t+1}-4$$
$$v_B\left(80\right)=50>49=v_B\left(80\right)$$

Rocket B is traveling faster at time $t = 80$ seconds.

8-2b

94.



Graph of v

A squirrel starts at building A at time $t = 0$ and travels along a straight, horizontal wire connected to building B. For $0 \le t \le 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

(a) At what times in the interval $0 < t < 18$, if any, does the squirrel change direction? Give a reason for your answer.

(b) At what time in the interval $0 \le t \le 18$ is the squirrel farthest from building A? How far from building A is the squirrel at that time?

(c) *Find the total distance the squirrel travels during the time interval* $0 \le t \le 18$.

(d) Write expressions for the squirrel's acceleration $a(t)$, velocity $v(t)$, and distance $x(t)$ from building A that are valid for the time interval $7 < t < 10$.

Part A

1 point is earned for the t-values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.

0	1	2

The student response earns all of the following points:

1 point is earned for the t-values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.

Part B

8-2b

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.

0	1	2

The student response earns all of the following points:

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.

Part C

8-2b

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)|dt = 140 + 50 + 25 = 215.$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)|dt = 140 + 50 + 25 = 215.$

Part D

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

For $7 < t < 10, a(t) = \frac{20-(-10)}{7-10} = -10$
$v(t) = 20 - 10(t - 7) = -10t + 90$
$x(7) = \frac{7+5}{2}.20 = 120$
$$x(t) = x(7) + \int_7^t (-10u + 90)du$$
$$= 120 + (-5u^2 + 90u)\Big|_{u=7}^{u=t}$$
$$= -5t^2 + 90t - 265$$

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for $a(t)$

8-2b

1 point is earned for $v(t)$

2 points are earned for $x(t)$

For $7 < t < 10, a(t) = \frac{20-(-10)}{7-10} = -10$

$v(t) = 20 - 10(t-7) = -10t + 90$

$x(7) = \frac{7+5}{2}.20 = 120$

$$x(t) = x(7) + \int_7^t (-10u + 90)du$$
$$= 120 + (-5u^2 + 90u)\Big|_{u=7}^{u=t}$$
$$= -5t^2 + 90t - 265$$

8-2b

95. A particle moves along the y-axis with velocity given by $v(t) = t \sin(t^2)$ for $t \geq 0$.

(a) In which direction (up or down) is the particle moving at time $t = 1.5$? Why?

(b) Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?

(c) Given that $y(t)$ is the position of the particle at time t and that $y(0) = 3$, find $y(2)$.

(d) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

1 point is earned for the answer and reason

$v(1.5) = 1.5 \sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

0	1

The student response earns all of the following points:

1 point is earned for the answer and reason

$v(1.5) = 1.5 \sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$
$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

0	1	2

The student response earns all of the following points:

1 point is earned for: $a(1.5)$

8-2b

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$
$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

Part C

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y\left(t\right) = -\frac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

$$\begin{aligned} y(t) &= \int v(t)dt \\ &= \int t \sin t^2 dt = -\frac{\cos t^2}{2} + C \\ y(0) &= 3 = -\tfrac{1}{2} + C \Rightarrow C = \tfrac{7}{2} \\ y(t) &= -\tfrac{1}{2}\cos t^2 + \tfrac{7}{2} \\ y(2) &= -\tfrac{1}{2}\cos 4 + \tfrac{7}{2} = 3.826 \text{ or } 3.827 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y\left(t\right) = -\frac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

8-2b

$y(t)=\int v(t)dt$

$=\int t\sin t^2 dt=-\frac{\cos t^2}{2}+C$

$y(0)=3=-\frac{1}{2}+C\Rightarrow C=\frac{7}{2}$

$y(t)=-\frac{1}{2}\cos t^2+\frac{7}{2}$

$y(2)=-\frac{1}{2}\cos 4+\frac{7}{2}=3.826\text{ or }3.827$

Part D

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

$\text{distance}=\int_0^2 |v(t)|dt=1.173$

or

$v(t)=t\sin t^2=0$

$t=0\text{ or }t=\sqrt{\pi}\approx 1.772$

$y(0)=3;\ y(\sqrt{\pi})=4;\ y(2)=3.826\text{ or }3.827$

$\left[y(\sqrt{\pi})-y(0)\right]+\left[y(\sqrt{\pi})-y(2)\right]$

$=1.173\text{ or }1.174$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

8-2b

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)|dt = 1.173$$

or

$v(t) = t\sin t^2 = 0$

$t = 0$ or $t = \sqrt{\pi} \approx 1.772$

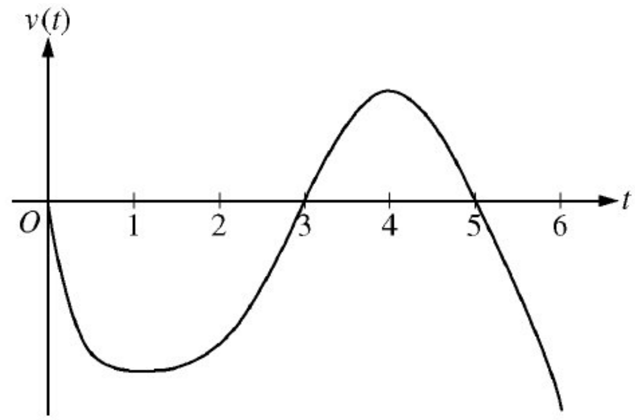
$y(0) = 3;\ y(\sqrt{\pi}) = 4;\ y(2) = 3.826$ or 3.827

$\left[y(\sqrt{\pi}) - y(0)\right] + \left[y(\sqrt{\pi}) - y(2)\right]$

$= 1.173$ or 1.174

8-2b

96.



Graph of v

A particle moves along the x-axis so that its velocity at time t, for $0 \le t \le 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t-axis and the graph of v on the intervals [0, 3], [3, 5], and [5, 6] are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

(a) For $0 \le t \le 6$, find both the time and the position of the particle when the particle is farthest to the left. Justify your answer.

(b) For how many values of t, where $0 \le t \le 6$, is the particle at $x = -8$? Explain your reasoning.

(c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.

(d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

The response can earn up to 3 points:

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t)\, dt$

$$x(3) = -2 + \int_0^3 v(t)\, dt = -2 - 8 = -10x(6) = -2 + \int_0^6 v(t)\, dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

0	1	2	3

8-2b

The response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t)\,dt$

$$x(3) = -2 + \int_0^3 v(t)\,dt = -2 - 8 = -10x(6) = -2 + \int_0^6 v(t)\,dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

The response can earn up to 3 points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

0	1	2	3

The response earns all of the following points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

8-2b

Part C

The response can earn up to 1 point:

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

0	1 ✓

The response earns all of the following point:

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

The response can earn up to 2 points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

0	1	2 ✓

The response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.